Accounting for Asset Pricing Factors

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Abstract. Many accounting numbers appear in standard factor models, but without a clear explanation. The numbers are generated under accounting principles that deal with risk, providing an explanation but also a critique of how extant models identify accounting-based factors. That leads to a revised factor construction. Accounting numbers are codetermined in a double-entry system, a feature that is exploited in packaging the factors into a model. Rather than entering as the separate, additive factors, adding to the "factor zoo", they are combined parsimoniously to capture the information they jointly convey about risk and return in the double-entry system. Empirical tests confirm.

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1. Introduction

Many factors in empirical asset pricing models are formed from accounting numbers, including book-to-price (B/P), investment, return on equity (ROE), and other profitability measures. These accounting numbers have been selected because they perform relatively well in explaining returns in the data. That hints of explanatory power but without explanation, particularly when the correlations just surface by data mining: Why do the accounting measures capture the risk to which the investor is exposed? Without that explanation, investors presumably add model risk to fundamental risk and that reduces informativeness for the rational Bayesian investor. Plausible conjectures are offered, typically equating accounting measures with economic constructs. Some factors are identified formally from economic theory, as in investment-based models, with accounting numbers then embraced as representing the economic constructs.

However, accounting numbers do not necessarily capture the constructs in mind; they are accounting phenomena, generated by accounting rules, and these rules depart from the economic ideal. A seemingly reasonable reaction is to adjust the measures for their so-called deficiencies, as several recent papers have done. However, there is method in the madness: Accounting principles that produce the so-called accounting distortions convey risk and return and this information can be exploited in building factor models. This paper does so.

The paper focuses on the B/P factor (called the value factor or HML in the Fama and French (1993) model and its subsequent extensions), the investment factor (the CMA factor in those extensions and a factor in the Investment CAPM of Hou, Xue, and Zhang, 2015 and its q^5 extension), and ROE and other profitability factors in these models. Each factor comes with

attributions in the literature: Distress risk, assets-in-place, risk in growth options, profitability connected to hurdle rates, cash-flow versus accrual profitability, and more. But each of the measures is determined by accounting principles that disturb them from the phenomena they are said to capture, and this accounting conveys information about the risk to investing. Recognizing this, the analysis here provides justification for some of the factors in these models but with a different explanation. However, it also shows that the standard factors do not fully incorporate the accounting information about risk and return, leading to alternative accounting-based factors.

The paper brings a further accounting feature to the construction of factor models. Extant models typically package identified factors together as separate, additive factors. With more factors successively "discovered" by data mining, this yields many potential factors, leading to the "factor zoo" complaint. The paper recognizes that accounting numbers are produced in a coordinating system, the double-entry system, under which the accounting for one measure affects another: A "debit" always requires a matched "credit." Accordingly, accounting numbers are co-determined, conveying information jointly, not as separate, additive pieces of information, and that can be exploited in building parsimonious factor models. The double-entry accounting system is itself a packaging system implying an alternative packaging of factors in a factor model, parsimoniously. We do so, producing a factor model that dominates most extant models in empirical tests. The double-entry system that has often been marveled at for its properties can also be utilized in constructing factor models, and the paper does so.

The paper is offered in the spirit of placing empirical asset pricing and the accounting on which it draws on the same platform. With a reported $2 trillion under management with factor investing, much is at stake.

2. Framing the Investigation

Two approaches have been taken in the construction of extant factor models.

First, in the mode of the Fama and French (1993) three factor model and its subsequent extensions in Fama and French (2015, 2018), potential factors are "discovered" by dredging the data for correlations that suggest return for risk. Modelling based only on observed correlations (without theory) is hazardous, but Fama and French (2015) bring an accounting-based pricing model to justify factors in their models. That is the familiar residual income model on which they conduct comparative statics to connect accounting features to the required return for equity investing. However, Penman and Zhang (2025) show that the comparative statics are inconsistent with how the double-entry accounting system works. That is the point of departure for this paper.

Second, the investment models of Hou, Xue, and Zhang (2015) and Hou, Mo, Xue, and Zhang (2021) are developed from production-investment theory and so are theoretically framed. The theory introduces economic concepts of investment, profitability, and growth in investment. However, for operationalizing the model, factors are formed on accounting measures that do not correspond to the concepts. For example, investment in the investment factor is that on the balance sheet but accounting omits much investment from the balance sheet, rather expensing it against earnings. That, in turn, affects investment in the denominator of the return-on-investment factor, and earnings in the numerator is reduced by investment that generates earnings. These accounting measures have an alternative risk interpretation to that imputed in investment models which we pursue.

The identification of factors here is developed under theory that ties accounting numbers to risk. Given the no-arbitrage dividend discount model with a constant discount rate, r, and clean surplus accounting under which earnings add to book value and dividends are paid out of book value,

$$P_t^T = \frac{\sum_{\tau=t+1}^{T}\left[Earnings_\tau + Dividend_\tau\left(\left(1+r^f\right)^{T-\tau}-1\right)\right]}{(1+r)^T-1} \rightarrow P_t, \quad (1)$$

the price in the infinite-horizon dividend discount model, as T → ∞ (with the usual "transversality condition" that dividends are ultimately paid out). (Asset subscripts are suppressed and variables subscripted τ > t are expected values, as at all points in the paper.) That is, price is expected cumulative future earnings with dividends reinvested (at the risk-free rate, r^f), capitalized at the required return. The derivation is in Ohlson (1995) who also shows that the expression is equivalent to an infinite-horizon residual income formula. The constant discount rate and risk-free rate are for simplicity, as is the model itself; the framing can be set up with a general asset pricing model with the risk discount represented by a varying covariance term, as in Penman and Zhang (2020).

Reverse engineering renders the required return (also called the expected return):

$$(1+r)^T - 1 = \frac{\sum_{\tau=t+1}^{T}\left[Earnings_\tau + Dividend_\tau\left(\left(1+r^f\right)^{T-\tau}-1\right)\right]}{P_t} \quad (2)$$

for T → ∞: The required return is given by expected future cum-dividend earnings relative to the current price that is discounted for the risk to the expected earnings. This is the expected earnings yield that Ball (1978) suggested is related to risk and return, but now formally so and with expected earnings over many periods as befits a going concern. Given no arbitrage,

$$(1+r)^T - 1 = \frac{P_T + \sum_{\tau=t+1}^{T} Dividends_\tau\left(\left(1+r^f\right)^{T-\tau}-1\right)-1}{P_t} \quad (3)$$

for T → ∞, so expected cum-dividend earnings in equation (2) yields the same discount rate that reconciles expected price plus dividend payoff to current price.

The theoretical and empirical question is what accounting numbers forecast this yield and thus can be a basis for construction a factor model. Information that forecasts earnings in the numerator of equation (2) is pertinent, but also information that conveys the risk discount to price in the denominator. However, apart from the clean-surplus relation, equations (1) and (2) are silent on the accounting for earnings. Indeed, random numbers for earnings satisfy the expression subject to the standard transversality condition whereby dividends must ultimately be paid out. More specificity is required to identify accounting that conveys the expected earnings yield and thus the required return. That is a matter of accounting principles.

2.1 Accounting Principles that Convey Risk

With apologies for being patronizing to the informed reader, this section lays out accounting principles and explains how they potentially convey risk and expected return.[1]

The governing principle for booking earnings involves an assessment of risk: Earnings are recognized only when uncertainty is resolved (fair value accounting aside). Until that "realization" point, the recognition of earnings is deferred. Thus, while a firm's market price will (rationally) incorporate future earnings expectations, the accounting informs that those earnings are still at risk. They have not yet been realized, so they should be discounted in the price that forecasts them. Earnings is revenue minus expenses, so this accounting is implemented under two sub-principles.

First, under the so-called revenue realization principle, revenue is not booked until a customer contracts, obligations under the contract are satisfied by both parties, and receipt of cash is

[1] Penman (2021) elaborates.

"highly certain."[2] Only then is revenue added to the income statement and, correspondingly, to book value, usually with a receivable or cash. Even then, a receivable is discounted (with an allowance for bad debts) to an expected cash equivalent. In asset pricing terms, revenue is recognized only when a firm can book a low-beta asset, cash or a near-cash (discounted) receivable. Until that point, revenue recognition is deferred as positive beta, an expectation at risk of not being realized—the customer might not materialize.

Second, the accountant then recognizes expenses incurred to gain revenues, so-called matched expenses, to report net value added from revenues. However, these matched expenses are disturbed by so-called conservative accounting that expenses some investments. While many investments, in inventory, plant, and equipment, for example, are booked to the balance sheet, others are charged against earnings if the outcome to the investment is particularly uncertain. R&D investment and advertising and promotion to build brand assets are the marque examples, but the accounting is pervasive, applied to investment to develop supply chains and distribution systems, customer loyalty programs, employee training and retention, software development, start-up and organization costs, and more.[3] So-called conditional conservatism—impairing the

[2] See the Financial Accounting Standards Board (FASB) ACS 606 and the International Accounting Standards Board (IASB) IFRS 15 standards on revenue recognition. These new standards replace earlier standards, but the basic principle remains, indeed reinforced.

[3] Under the IASB Conceptual Framework, an uncertainty criterion is applied for an asset to be booked to the balance sheet. For the accounting for R&D under FASB Statement No. 2, the FASB requires the investment to be expensed due to the "uncertainty of future benefits." In IAS 38, the IASB applies the criterion of "probable future economic benefits" to distinguish between "research" (which is expensed) and "development" (which is capitalized in the balance sheet and amortized). Even IAS 16 on property, plant, and equipment requires benefits to be "probable" for the asset to be booked. This does not imply that the formal accounting requirements always "get it right" but to demonstrate the application of the principle in practice. While these principles are in regulatory accounting standards, the practice of conservative accounting largely developed voluntarily before twentieth-century regulation, as did the realization principle.

assets actually booked to the balance sheet on lowered expectations of payoffs—reinforces this accounting.

The risk that this accounting responds to might be idiosyncratic risk, nothing to do with the priced risk that asset pricing theory models. However, Penman and Zhang (2020) show that the two principles—the revenue realization principle and conservative accounting for investment—inform about the covariance discount factor in a general, no-arbitrage asset pricing model. That theory guides our identification of accounting-based factors. Empirical support is in Penman and Reggiani (2018), Penman and Zhang (2021), and Penman and Yehuda (2019).[4]

3. Identifying Accounting-based Factors: Empirical Analysis

The empirical analysis is in two parts. In this section, factors identified under accounting theory are evaluated relative to those in extant factor models on how they convey expected returns. Then, in Sections 4 and 5, identified factors are packaged into a parsimonious factor model under the governance of the double-entry system and compared empirically to extant factor models.

The empirical analysis in each subsection in this section covers a specific accounting number appearing in extant factor models. We begin each explaining how the number conveys risk under accounting principles (or not) with alternative numbers that do so then identified. The empirical analysis tests whether the numbers are priced in accordance with the underlying accounting principles. In the mode of empirical asset pricing research, this is in two parts. The first involves cross-sectional "characteristic regressions" that report how the measures correlate with forward stock returns. These are the regressions in data mining for factors, but now as tests of predictions for coefficient estimates based on accounting principles. As pointed out in Fama and French

[4] The idea of tying the expected return to the timing of risk resolution is not foreign to asset pricing research. See, for example, Epstein and Zin (1989), Bansal and Yaron (2004), Dechow, Sloan, and Soliman (2004), Weber (2018), and Mullins (2020).

(2020), estimated slope coefficients can be interpreted as factor returns. In the second part, identified accounting numbers from the regressions are formed into "factor mimicking" portfolios, potential factors to be included in a factor model. The progression from characteristic regressions to factor construction follows the development of the three-factor model in Fama and French (1993) from their observation of characteristics in Fama and French (1992).

The standard caveat applies to our empirical analysis: We assume that the market prices risk efficiently, as is standard in empirical asset pricing. The accounting-arbitrage alternative—investors do not process accounting information efficiently—cannot be taken off the table. Note also that many t-statistics are reported, so the reader is reminded of the chance of reporting significant statistics with repeated drawings from the t-distribution even though there is no substantive relation. However, the analysis is based on accounting principles connected to risk in theory, mitigating both concerns.

Compustat and CRSP are the data sources. Annual accounting measures are calculated for each year, 1964-2018, for all firms with the required data on Compustat in that year, except for financial firms (SIC codes 6000-6999) and utilities. Firms classified as Membership Organization or Unknown SIC are also excluded. Forward annual returns about which the accounting informs cover the twelve months beginning three months after fiscal-year end by which time the relevant accounting data must be reported, by law. So, for December 31 fiscal-year firms, returns run through March 2020. The return period for each firm covers the period when (realized) accounting numbers in the expectation in price at the beginning of the return period are reported (in four quarterly reports). The appendix details the calculation of variables.

Table 1 reports summary statistics for measures that enter later in the paper. Panel A summarizes their distributions and Panel B average cross-sectional correlations for selected

measures. We simply post these tables at this point, with commentary deferred until the measures are introduced in the empirical analysis. The correlations in Panel B are important when asking if alternative measures are capturing the same thing. As factor portfolios are formed by ranking variables, we will be referring to the rank correlations.

3.1 Accounting for Investment

An investment factor appears in both Fama and French (FF) models and investment CAPM models. The investment number is the change in balance-sheet assets. However, this number is not the firm's investment but rather just the part of that investment that accountants book to the balance sheet. Many investments are expensed against earnings under the conservative accounting principle of Section 2.1. This principle distinguishes the two treatments based on risk: Investments where the outcome is particularly uncertain are not booked as assets.

Considerable research, in Titman, Wei, and Xie (2004), Cooper, Gulen, and Schill (2008), and Fama and French (2006), for example, reports that balance-sheet investment yields lower returns. In the parlance of finance, investment on the balance sheet is the realization of expected investment opportunities, a revision of the risk that investment opportunities will not be realized; it thus accords with intertemporal asset pricing theory in Merton (1973) under which risk is to (unrealized) future investment opportunities. Thus, resolving that risk favorably means lower expected returns. In Berk, Green, and Naik (1999), risk declines as firms convert risky growth options to "assets in place", and that is recognized in the accounting treatment.

However, in accordance with the accounting principle, empirical tests indicate that expensed investments are priced to yield stock returns commensurate with higher risk. Lev and Sougiannis (1996) report higher returns to R&D investment though they choose to interpret this as market

inefficiency. Oh and Penman (2024) report that R&D investment is priced in the stock market with a higher risk discount than investment booked to the balance sheet and Oswald, Simpson, and Zarowin (2020) report that R&D that is capitalized to the balance sheet under IFRS when it is demonstrated to be successful is priced at a two percent premium to R&D that is expensed as not yet successful.

Characteristic regressions

Panel A of Table 2 reports the results from annual cross-sectional regressions of forward returns, Ret_{it+1} in excess of the risk-free rate, on investment measures. As in all cross-sectional regressions in the paper, reported coefficients are means over years, with t-statistics (in parentheses) calculated as those mean coefficients relative to their standard errors estimated from the time-series of coefficient estimates (Fama-and-Macbeth style). As factors are formed from a ranking of variables, the results are from decile-rank regressions, but those with continuous variables with an outlier treatment are similar.

Regression 1 in Panel A regresses forward returns in the investment measure in the Fama and French (2015 and 2018) five- and six-factor models, the Hou, Xue, and Zhang (2015) Investment CAPM, and the Hou, Mo, Xue, and Zhang (2021) q^5-factor model where $\text{Investment}_t/\text{TA}_{t-1} = \Delta\text{Total Assets}_t/\text{Total Assets}_{t-1}$. The mean coefficient on Investment_t is reliably negative, consistent with previous findings and intertemporal asset pricing theory and also consistent with the realization principle in accounting: Investment booked to the balance sheet indicates a higher probability of realizing earnings, thus conveying lower expected returns.

Regression 2 with expensed investment reports a different result. Here $\text{ExpInv}_t/\text{TA}_{t-1} = (\text{R\&D}_t + \text{Advertising}_t + (\text{SG\&A} - \text{Advertising})_t \times 0.3)/\text{Total Assets}_{t-1}$ which has a mean correlation with

$Investment_t/TA_{t-1}$ of only 0.16.[5] An estimated portion of SG&A (Selling, General, and Administrative Expense) is included as expensed investments charged to that account, as in Peters and Taylor (2017). In contrast to the negative mean coefficient on $Investment_t/TA_{t-1}$ in Regression 1, that on expensed investment is not significantly different from zero. This investment is priced differently: The t-statistic on the mean difference (over years) of -0.016 between the coefficients in Regressions 1 and 2 is -3.04.[6] The 30 percent of SG&A is presumably a very noisy estimate, but results are similar with the SG&A component excluded.

The differences in slope coefficients are accentuated when $Investment_t$ and $ExpInv_t$ are denominated relative to P_{t-1} in Regression 3. Price incorporates expectations so the investments are relative to expectations and in that price. Further, the measures have the same (lagged) price as returns, so additions to price in returns are compared to additions to price with accounting numbers, well-specified. And price is the deflator in the framing equation (2) that incorporates the discount for risk to expected earnings. With this deflator, the mean coefficient on $Investment_t$ is -0.009 (t = -2.95) and that on $ExpInv_t$ is 0.012 (t = 4.95).[7] The t-statistic on the mean difference (over years) between the two coefficients is -5.00.

[5] In the correlation panel in Table 1, the investment numbers are denominated in price for later reference. The 0.16 here refers to the denomination in lagged assets, as in the table here. $R\&D_t + Advertising_t + (SG\&A - Advertising)_t$ x 0.3) is 15.9% of lagged total assets on average.

[6] The investment measure increases with realized earnings reinvested in the business. So, in part, it captures current earnings realizations (and fully so with no dividend payout or net share issue). Similar results are observed when investment is confined to changes in gross property, plant, and equipment and inventory (tangible assets), as in some papers. Results are also similar with $Investment_t = \Delta NOA_t / NOA_{t-1}$ (that excludes investment in cash) and with expensed investment, $ExpInv_t$, denominated in NOA_{t-1}. NOA is Net Operating Assets = Operating Assets – Operating Liabilities (defined in the appendix). Some papers, for example, Cho, Kremens, Lee, and Polk (2021) define investment as change in book equity or net share issuance, but equity can change with a debt-equity swap that does not affect investment in the business (as with a repurchase financed by borrowing).

[7] With $ExpInv_t/P_{t-1}$ in the regression alone, as in Regression 2, the mean slope coefficient is 0.014 (t = 5.20).

In both the Fama and French (FF) extended factor models and the *q*-investment models, investment enters with a profitability measure. In the latter, profitability, measured by ROE, is implied by *q*-theory as the relation between investment and the required return is conditional on expected profitability. So Regression 4 adds ROE, as in the investment models, and Regression 5 adds the cash-flow profitability measure of Ball, Gerakos, Linnainmaa, and Nikolaev (2018), CP_t/TA_{t-1} chosen in Fama and French (2018). These profitability measures add explanatory power (for reasons later), but mean coefficients on the investment numbers are little different; the difference in coefficients on balance-sheet $Investment_t$ and expensed investment $ExpInv_t$ remain.

The final regression in Table 2 adds characteristics underlying factors in the Fama and French six-factor model, market beta, B/P, Size, and Momentum (MOM) to Regressions 5. The mean coefficient on expensed investment remains significantly positive: While investment booked to the balance sheet is included in these models (with a negative association with returns, as here), expensed investment incrementally projects higher average returns. These models do not capture this aspect of risk and return relevant to asset pricing. However, B/P and Size add to the explanation of returns, promoting the investigation in subsequent subsections.

Investment Factor Returns

With a view to factor model construction, Panel B of Table 2 reports mean returns from long minus short positions determined from ranking firms on investment measures each year. The difference in returns between the top 1/3 of stocks (High) and the lowest 1/3 (Low) is interpreted as the return to a "factor mimicking" portfolio, as in the factor construction in most extant models. To align in calendar time, the long and short positions are determined from a ranking of

firms at March 31 each year, with firms entering the ranking based on the accounting measure for its most recent fiscal year.[8]

The $Investment_t/TA_{t-1}$ factor is the CMA in FF models and IA in *q* models. For both this and $ExpInv_t/TA_{t-1}$, the factor returns are those implied by the regression results, -0.093 for balance-sheet investment but an insignificant 0.025 for expensed investment. However, the panel reports a very different result with the two investment numbers relative to P_{t-1}. The negative factor return for $Investment_t$ survives, but the 0.105 return for $ExpInv_t$ is now significantly positive.[9]

The difference in the results with the two deflators is explained by Total Assets/Price, part of Book/Price, and that depends, in part, on the amount of investment expensed instead of booked to book value. B/P is ranked correlated 0.33 with $ExpInv_t/P_{t-1}$ in Table 1 but the correlation with $ExpInv_t/TA_{t-1}$ (not reported there) is -0.26. That leads us to the next section.

3.2 Accounting for Book Value

Book-to-price is ubiquitous in asset pricing, with higher B/P said to indicate higher risk and thus the basis for a B/P factor. Book value is often interpreted as "assets in place", with B/P indicating the value (and risk) of "growth opportunities over assets in place." However, it is accounting number determined by how book value is measured

Characteristic regressions

[8] The March 31 date means the ranking number is recent for the majority of firms (with December 31 fiscal-years), but it is staler for fiscal-years ending in other months. Results are similar with the ranking at each June 30, as in the Fama and French construction.

[9] Bongaerts, Kang, and van Dijk (2021) calculate an intangible asset factor based on expensed investment and report its associated Sharpe Ratio is higher than that for established investment factors.

Table 3 has a similar layout to Table 2, but with book value measures rather than investment. Regression 1 in Panel A reports the standard finding: B/P is positively related to returns. The measure in Regression 2, $\text{CapExpInv}_t/P_t = \frac{Capitalizec\ Expensed\ Investment_t}{P_t}$, capitalizes the expensed investment in Table 2 with ex ante amortization rates in the variable description in the appendix.[10] It is denominated in the same price as B/P. Panel A of Table 1 informs that capitalizing expensed investment adds significantly to balance sheet assets, 31.6 cents per dollar of price at the mean and 13.9 cents at the median with wide cross-sectional variation. Panel B there reports a mean rank correlation with B/P of 0.46, and the mean coefficient on this as-if book value in Regression 2 is similar to that on B/P in Regression 1. Results are similar when the capitalized SG&A component is excluded.

The incremental explanation in capitalized expensed investment is clear in Regressions 3 and 4: Given B/P, the capitalized expensed investment adds to the explanation of returns. These regressions include the profitability measures, not only because these measures are included with B/P in factor models, but because, under double entry, book value cannot be considered without the complementary earnings which the accounting for book value affects. Regressions 5 and 6 show that this holds with added characteristics that enter existing factor models. With the exception of Size_t, these characteristics do not add explanatory power.

[10] An ex ante amortization schedule for an asset where outcomes are highly uncertain and inevitably contains error. Amortization rates are often assumed or estimated from data that involves only surviving firms where the investment was successful, as in Lev and Srivastava (2020), for example. With this ex post bias, these rates are not applicable ex ante where less successful outcomes are also entertained; amortization rates based on successful outcomes would result in anticipated impairments. See the commentary in Penman (2025). Peters and Taylor (2017) use BEA industry-specific amortization rates for R&D and a constant 20% for the capitalized SG&A component. Arnott, Harvey, Kalesnik, and Linnainmaa (2021) follow the Peter's and Taylor (2017) scheme. We make no claim that our assumed rates are any better. Barker, Lennard, Penman, and Teixeira (2021) discuss the accounting issues in recognizing these "intangible" assets. Note that we ignore deferred taxes which likely would be booked if an accountant capitalized expensed investments in practice. (These presumably would also be discounted in price as uncertain tax provisions.)

The findings for capitalizing expensed investment are consistent with those in several papers that add expensed investment to book value in forming a B/P factor.[11] However, pooling assets of different risk loses information, as accounting principles and the empirical results indicate. This is the issue of how accounting numbers are to be packaged into a factor number, which we take up in Section 4.

B/P Factor Returns

Panel B of Table 3 reports returns for factor portfolios formed from B/P and capitalized expensed investment, constructed in the same way as in the corresponding panel in Table 2. The High – Low (factor) annual return for capitalized expensed investment (0.116) is higher than that for B/P (0.091), though the t-statistic on mean difference between the two is only1.08. For the sum of the two book value measures in $TotB_t/P_t$, the High - Low return of 0.106 is only a little higher than that for B/P (the t-statistic on the mean difference is 1.78).

Another accounting feature bears on book value in factor models: With double entry accounting, book value and earnings are jointly determined, so book value cannot be considered without the corresponding earnings. That is clear from the accounting that omits investments from the balance sheet—it requires them to be expensed to the income statement. The original Fama and French (1993) model did not include earnings, but earnings enter the extended FF models via ROE. That leads to the next section which points to an alternative factor design involving both book value and earnings.

3.3 Accounting for ROE

[11] See Li (2020), Park (2022), Lev and Srivastava (2020), Kapons (2020), Arnott, Harvey, Kalesnik, and Linnainmaa (2021), Eisfeldt, Kim, and Papanikolaou (2020), and Eisfeldt and Papanikolaou (2013).

In existing factor models, ROE is interpreted as a measure of return on investment, with *q* theory then justifying its inclusion. But the accounting for ROE gives it a different interpretation. An economic measure of return on investment would compare the earnings from investment to an investment base. With conservative accounting for investment, earnings in the numerator of ROE is reduced by expensed investment, mixing investment with return on investment, a seeming perversion. Consequently, this investment is missing from the denominator. Feltham and Ohlson (1995) and Zhang (2000) demonstrate these deterministic accounting features. With these seeming distortions, conjectures that a higher ROE results from a higher hurdle rate (and thus indicates higher risk) are off base, as are investment CAPM models that embrace ROE as measuring the productivity of investment.

However, the conservative accounting for investment that affects ROE conveys risk. A firm with low earnings (and thus a low ROE) due to expensed investment is potentially a high-risk firm—it is making investments that may not pay off. If earnings on the investment are realized, ROE is high because, due to the initial expensing, the investment is not in the book value base. Thus, a high ROE indicates risk realized and a lower expected return for risk. This contrasts with existing factor models where ROE is said to be positively related to risk. Penman and Zhang (2020, 2021) provide the theory and supporting empirical analysis.

However, there are subtleties involved. First, under the deterministic leveraging equation,

$$ROE_t = RNOA_t + Leverage_{t-1}[RNOA_t - \ Net\ Borrowing\ Rate_t],$$

where $RNOA_t$ is the book return on net operating assets (the return on business operations, the enterprise return) and $Leverage_{t-1} = Net\ Debt_{t-1}/Equity_{t-1}$. So, as leverage adds to risk and the expected equity return in theory, higher ROE indicates a higher expected return to the extent it is

affected by leverage. The conservative accounting effect enters through RNOA in the opposite direction, with expensed investment reducing RNOA. So it is a question of which dominates.

Second, in many asset pricing models, ROE enters along with B/P, with ROE_t taken as an expectation of forward ROE at risk. However,

$$ROE_{t+1} \times \frac{B_t}{P_t} = \frac{Earnings_{t+1}}{B_t} \times \frac{B_t}{P_t} = \frac{Earnings_{t+1}}{P_t}.$$

That is, adding forward ROE to B/P recovers the forward E/P ratio (multiplicatively), and does so additively with the logged variables in most characteristic regressions in asset pricing research. The forward E/P ratio gains its legitimacy as an indicator of the expected return from equation (2), consistent with the yield interpretation of E/P in Ball (1978). In support, E/P is positively associated with returns in many empirical tests, as in Basu (1977, 1983). In most asset pricing models, this joint implication of B/P and ROE is not recognized; rather, the two numbers enter as separate contributors to explaining returns and the basis for separate, additive factors. But, instead, B/P implicitly drops out in the construction, with the focus effectively on E/P: Under the accounting principle whereby earnings and book values articulate in a double-entry system, the product, $ROE_{t+1} = \frac{Earnings_{t+1}}{B_t} = \frac{Earnings_{t+1}}{P_t} \times \frac{P_t}{B_t}$, is insensitive to how book value is measured: lower book value implies higher $\frac{Earnings_{t+1}}{B_t}$ by accounting construction without an effect on E/P.

Third, in tying ROE to expected returns, there are differences between positive and negative ROE. By the leveraging equation above, leverage decreases ROE for negative RNOA providing the borrowing rate is positive. Further, while ROE is typically positively serially correlated,

negative current ROE_t has been documented to be less informative about future ROE—it is affected by transitory items.

Characteristic regressions

Regressions in Table 4, Panel A are with positive ROE, Panel B with negative ROE. In contrast to the positive association of ROE with returns (conditional on B/P) typically reported, Regression 1 reports a slightly negative correlation in Panel A (unconditionally). For negative ROE in Panel B, the mean coefficient is positive, indicating that less negative ROE projects higher future earnings in equation (2), distinguishing temporary losses from those expected to be more permanent. Regression 2 adds B/P which loads with a positive coefficient, and ROE is now positively associated with returns in both panels (conditionally). This is the characteristic regression result that has prompted the addition of an ROE factor to a B/P factor in factor models, with the inference that higher ROE conveys higher risk and return. On the face of it, that seems strange: Profitable firms are more risky?

However, by the ROE x B/P = E/P calculation, ROE with B/P implies E/P, an indicator of the (future) earnings yield on current price. Regression 3 thus has E/P alone, with a significant positive correlation for profit firms: E/P supplies information in B/P and ROE. That fits with the framing equation (2): Current earnings denominated in price indicates expected future earnings denominated in the same price (discount), suggesting that E/P conveys *r*. That point is strengthened with by the observation that trailing P/E ratios and forward P/E ratios are strongly positively correlated. And it suggests an E/P factor rather than the two separate, additive B/P and ROE factors. The mean negative coefficient for loss firms also projects higher returns (with the negative E/P).

Regression 4 adds ROE to E/P rather than to B/P. If ROE is depressed by conservative accounting for investment, it conveys risk and return under accounting principles and thus is a commentary on the earnings in E/P. For profit firms, the mean coefficient on ROE is negative; a lower (higher) ROE for a given E/P indicates higher (lower) returns. As ROE and E/P have the same numerator, the only number added in Regression 4 over Regression 3 is book value. But rather than entering in B/P, book value enters as the denominator of earnings in ROE, and that conveys risk under conservative accounting: As that accounting results in lower book value when investment is expensed, realized earnings on low book value indicates lower risk—the investment omitted from book value as particularly risky has paid off. For loss firms, the mean coefficient on earnings is negative, projecting higher returns for higher losses, but that on ROE is positive, indicating that, given negative E/P, higher (less negative) ROE project higher future earnings in equation (2) with lower risk.

Regression 5 adds B/P to Regression 4, to assess whether it now adds to explaining returns. B/P loads for profit firms but ROE now adds little explanatory power. This is explained by B/P = E/P x 1/ROE. That is, given E/P, adding B/P is equivalent to adding the inverse of ROE. In short, book value conveys information about risk and return in conjunction earnings, not as a separate feature relative to price but by information in the denominator of ROE. For loss firms, B/P also takes from the coefficient on ROE. This not only revises the standard interpretation of ROE in extant models, but also explains why B/P enters.[12]

[12] With current ROE_t in the regression, the relation ROE x B/P = E/P is an approximation. Precisely, $\frac{Earnings_t}{B_{t-1}} \times \frac{B_t}{P_t} = \frac{Earnings_t}{P_t} \times \frac{B_t}{B_{t-1}}$. Adding $\frac{\Delta B_t}{B_t}$ to Regression 4 does not change the inferences.

A low ROE in the cross-section can be due to conservative accounting that expenses particularly risky investment, and that is the attribution given to coefficients on ROE in the regressions here. However, a low ROE can also be due to poor ex post performance. So, Regression 6 adds the expensed investment which is the conservative accounting effect on the numerator of ROE. The two are negatively correlated, -0.23 on average. The $ExpInv_t/P_t$ measure loads with a positive mean coefficient for both profit and loss firms, as expected of a variable that conveys risk. The expensed investment variable takes away some of the explanatory power of ROE for profit firms, reinforcing the attribution to the conservative accounting effect on ROE. For loss firms, the ROE coefficient remains positive, supporting the conjecture that ROE in this case conveys information of the extent to which losses are transitory.

Regression 7 adds standard factor characteristics, Beta, Size, and Momentum to Regression 6. With the exception of Size, these variables do not load, although R^2 increases. The mean coefficients on ROE remain similar to those in Regression 6. Regression 8 involves ROE, Investment, and Size which together enter *q*-investment models. The mean coefficient on ROE is positive, consistent with *q*-theory that predicts that, controlling for investment, ROE indicates higher risk and return. However, with the addition of E/P and expensed investment in Regression 9, the coefficient on ROE goes to zero for profit firms. The positive coefficient on ROE survives only with loss firms (negative ROE) in Panel B where Investment is no longer significant.

Factor Returns

Panel C of Table 4 reports potential factor returns based on the pertinent variables in the regressions in Panels A and B. The long minus short return for positive E/P is positive, consistent with the regressions, though not so for negative E/P. In contrast, the corresponding return for ROE, without the conditioning B/P, is not significantly different from zero. The mean factor

return for $ExpInv_t/P_t$ is a significant 11.7%. So, while a higher E/P indicates a higher expected return on average, higher expensed investment that reduces E/P conveys a higher expected return. It follows that, in projecting the earnings yield for the required return in equation (2), $ExpInv_t/P_t$ must be accommodated, and that will be done with the packaging of accounting measures into a factor model in Section 4.

3.4 Accounting for Alternative Profitability Measures

FF models involve alternative profitability measures to ROE. Table 5 adds the operating profitability measure, OP_t/B_{t-1}, and the cash-flow profitability measure, CP_t/TA_{t-1}, to selected regressions in Table 4 to see if they add information about expected returns to that in ROE and the other accounting measures there. The comparison with ROE_t is intriguing because the measures exclude expensed R&D, though include SG&A which may involve expensed investment. The mean cross-sectional rank correlation between CP_t/TA_{t-1} and ROE_t in Table 1 is 0.53, and 0.86 for OP_t/B_{t-1}.

With this positive correlation, the operating profitability measure in Regression 1 in Table 5 returns similar coefficients to ROE in Table 4 in both panels. Further, OP_t/B_{t-1} adds little explanatory power to E/P and ROE in Regression 4, and the positive coefficient on E/P and negative coefficient on ROE are little changed from that in the corresponding regression in Table 4. Moving across the panels to Regression 7 (that includes expensed investment) and Regression 10 (with other Fama and French features), OP has little additional explanatory power for profit firms.

Not so for the cash-flow profitability measure. In Regression 2, 5, and 8, it adds positively to returns in both panels, with and without ROE and expensed investment in the regressions.

Further, in Regression 11 it adds to the other features in FF models, supporting it as the choice of profitability measure in Fama and French (2018) as fitting best to returns. We also investigated the Novy-Marx (2013) gross profitability measure. Like CP_t, that loaded with significant positive coefficients in regressions similar to Regressions 5 and 8, but not in Regression 11.

As these profitability measures exclude R&D, we suspect they might refine E/P as an expectation of the future earnings yield in equation (2).[13] So, Regression 12 drops E/P from Regression 5, leaving CP_t profitability and ROE_t. Cash-flow profitability takes up the positive coefficient on E/P for profit firms while that on ROE remains negative, though not for loss firms. This survives with $ExpInv_t/P_t$ added in Regression 13. Regression 14 denominates CP_t in P_t rather than TA_{t-1}, a measure of yield to price, and the coefficient on CP_t/P_t is 0.015 with a t-statistic of 5.99 for profit firms and also significantly positive for loss firms. With expensed investment still in the regression, that is telling. Perhaps CP_t/P_t is a yield that is closer to a yield from investment than E/P and accordingly is a better indicator of the expected earnings yield in equation (2). Yet why the exclusion of short-term accruals (but not long-term accruals) would contribute is unclear.[14]

[13] To be frank, we do not quite understand what OP_t and CP_t are capturing, except to report that they are not capturing the risk and return conveyed by conservative accounting. CP_t is nominated as a cash flow measure. It adjusts operating earnings for short-term accruals but not depreciation and amortization (accruals) and long-term accruals such as those for pensions, long-term receivables, and compensation. (Long-term deferred revenues are taken out). By excluding accruals it omits aspects of earnings realizations—sales on credit, expenses incurred but not yet paid for, and deferred revenues not yet realized, for example. Gross profitability is clearer, the immediate profit from revenues from trading products and services over their costs. But leaving out other expenses to generate this revenue is at odds with a full accounting for earnings from revenues; these are not clean-surplus earnings. OP_t includes interest, very much like ROE. Neither of the measures include taxes but effective tax rates vary considerably with tax planning in the cross-section.

[14] Ball, Gerakos, Linnainmaa, and Nikoleav (2020) introduce retained earnings relative to price, showing that it forecasts returns better than B/P. This measure of accumulated (average) past earnings could be a better predictor of the future earnings yield than a one period of earnings yield, but it did not perform well in regressions relative to the profitability metrics in Table 5. This might be expected given the metric reduces past earnings for payout whereas the target is equation (2) is cum dividend.

Regression 3 introduces another profitability measure,

$$\mathrm{ROE}_t^{\mathrm{Adj}} = \frac{(\mathrm{Earnings}+\mathrm{Expensed\ Investment}-\mathrm{Amortized\ CapExpInv})_t}{\mathrm{TotB}_{t-1}}.$$

This measure reverses the conservative accounting effect on both earnings and book value to estimate a book rate of return unperturbed by the accounting. The idea here is to develop a book rate of return that is closer to the economic profitability measure envisioned in the investment CAPM. The denominator a book value that includes capitalized expensed investment and the numerator replaces expensed investment in earnings with the amortization of their capitalized value. For profit firms, the correlation with returns is negative, zero for loss firms. It adds little when added in the other regressions, indicating that "correcting" the accounting does not produce a profitability measure that better explain returns.

Factor Returns

Panel C of Table 5 repeats the portfolio construction in the corresponding panel in Table 4 but now with the alternative earnings measures relative to price, cash flow profit yield, CP_t/P_t, and operating profit yield, OP_t/P_t. Both measures report significant long minus short returns in the panel, higher that those with E/P in Table 4, though the difference is negative for negative profits.

3.5 Size?

In Tables 2 - 5, firm size adds explanatory power. The "size effect in returns" is not well explained in the literature, at least in terms of fundamentals. Size is not an accounting number, but could it be associated with risk to expected earnings? That makes sense: Small firms are

those investing intensively in R&D, advertising, software, and other expensed intangibles for which earnings are not yet realized and are at relatively high risk under accounting principles. The $ExpInv_t$ measure partly captures this, but size might proxy more widely for expected earnings at risk. In contrast, large firms—a mature pharmaceutical rather than an early-stage biotech, for example—are those realizing earnings and thus less risky. This raises the specter that the size premium in returns is explained by the extent of earnings at risk, a hypothesis investigated in the affirmative in Penman, Reggiani, Richardson, and Tuna (2018) and Penman and Reggiani (2018, 2024).

We partitioned positive ROE firms each year into large (highest 1/3 in a size ranking) and small (lowest 1/3) and found that, for Regression 4 in Table 4, the mean coefficient on ROE for small firms is -0.021 (t = -3.93) while that for large firms is only -0.003 (t = -0.90).[15]

Many papers exclude micro-caps when constructing factors. So we re-estimated Table 4 excluding firms below the 20th percentile of NYSE firms ranked on market capitalization. For loss firms, results were similar. For profit firms, the mean coefficient on ROE in Regression 4 was only -0.003 (t = -1.06) but all other results were similar. This reinforces that the accounting effect is stronger in smaller firms. The next section carries the investigation further.

4. Packaging Accounting Measures into a Factor Model

By recognizing accounting principles that pertain to risk, the preceding analysis has identified potential factors that imbed those principles. However, the investigation that critiques existing

[15] This points to a resolution of a puzzle in Fama and French (2015), p. 16: "The biggest problem for the five-factor model in Table 11 is the portfolio of small stocks in the lowest profitability and highest investment quartiles."

factor models has added the to the factor zoo. How should a factor model be built from these observations in a parsimonious way?

In most extant models, each factor enters as a separate, additive feature said to add to the explanation of priced risk.[16] Is this the way to construct factor models? The principles of accounting say: No. Accounting numbers are produced jointly in the double-entry system, not as independent features, so they jointly convey risk and return under accounting principles. Thus, book value does not enter as a stand-alone measure potentially conveying risk, but with the earnings with which it is co-determined (as above). And ROE enters, not as a basis for an additional independent factor but as a measure involving both earnings and book value that are jointly determined, conveying risk and return together (as the characteristic regressions confirm). This calls for a different packaging of the relevant accounting numbers into a factor model, to which we proceed. There is no claim that we are producing the definitive model; we are just bringing evidence of how accounting principles applied in a double-entry system can be exploited to build factor models.

4.1 SUMIS: An Income Statement Factor Based on Earnings, ROE, and Expensed Investment

Current earnings-to-price, E/P is denominated in the same price as in equation (2), so is a potential indicator of expected earnings-to-price and of *r*, and the characteristic regressions confirm. Indeed, if current earnings were a sufficient statistic for expected future earnings in the numerator of equation (2), the matter would end there (and the E/P factor return in Panel C of Table 4 would say it all). However, current earnings are reduced by expensed investment that produces the expected future earnings, and that investment conveys risk under conservative

[16] Factor interactions are involved in Duan, Gong, and Qi (2021). In *q*-investment models, the investment and profitability factors are conditional on each other.

accounting. And, under double entry, the book value affected by the conservative accounting is also relevant, for it is the denominator of ROE.

This suggests factor construction along the following lines. With E/P as a starting point, distinguish low versus high ROE for a given E/P. That incorporates effect of conservative accounting on the earnings in E/P (via the ROE numerator) but also the effect on book value (via the denominator); a high ROE indicates earnings realized on previously expensed investment that produced the low book value—risk resolved—and a low ROE indicates risky expensed investment with future earnings still at risk, yet to be realized. But a low ROE can also be due to poor ex post performance. So, within those ROE groups, identify the extent to which ROE is affected by the expensing of investment. Regressions 4 and 6 in Table 4 so inform.

So, each year at March 31, firms are sorted into four portfolios, one with all negative E/P and three from a ranking on positive E/P, low, medium, high, 1/3 each. Then, within each of those portfolios, firms are sorted on ROE, with the top and bottom third designated high and low. Finally, within each of those E/P-ROE portfolios, firms are sorted on $ExpInv_t/P_t$ with the top and bottom third again designated high and low. Importantly, these are nested sorts to capture the joint accounting effects for the same set of firms, not the independent sorts of extant models.

Panel A of Table 6 reports mean annual excess returns over years for portfolios formed from the first two sorts, on E/P and then ROE. The returns are increasing in E/P (across rows), though those for negative E/P are higher than the low positive E/P. For a given E/P, returns are decreasing in ROE (down columns); ROE adds to information about expected returns to that in E/P, with the difference between high ROE in row 1 and low ROE in row 3 statistically significant for all positive E/P. Beginning-of-year book value, the denominator of ROE, is the added information is this second sort; a higher (lower) book value (due to conservative

accounting) relative to the earnings in the first sort indicates higher (lower) returns.[17] The panel reports (at the far right) the mean return difference between the high positive E/P portfolio each year with low ROE (superscripted a) and the low positive E/P portfolio with high ROE (superscripted b), the return difference across the diagonal spread in the table. The mean return spread (with zero net investment) is a significant 7.6%, indicating payoffs to risk differences in the portfolios.

Implicitly, the portfolio formation involves B/P: Because B/P = E/P x 1/ROE (approximately), a ranking on ROE is an inverse ranking on B/P for a given E/P. (The book value is beginning book value but B/P ratios are strongly serially correlated.) Thus B/P, so ubiquitous in asset pricing models, enters, but in a quite different way. Double entry (whereby earnings and book articulate) requires earnings and book value to enter jointly, as here.

Panel B of the table adds the third sort on expensed investment, $ExpInv_t/P_t$, to that in Panel A. Returns in row 1 are for high ROE but with low $ExpInv_t/P_t$, deemed relatively low risk by accounting principle, and those in row 3 for low ROE but high $ExpInv_t/P_t$, deemed high risk. With realized earnings, book value, and expensed investment involved, both the realization principle and conservative investment principle are incorporated with both income statement and balance sheet effects. The return difference between the two is reported as rows 3 – 1. The t-statistics indicate that the differences are significant, enhanced from Panel A, with those for negative E/P now positive (but not large).

[17] Tests confirm that the ranking on ROE is not just a further ranking on E/P. Leverage adds to risk and return and to ROE, so the negative correlation between ROE and returns cannot be attributed to the leverage effect on ROE. But note that leverage (risk) is built in: Leverage increases both E/P and ROE. See Penman, Reggiani, Richardson, and Tuna (2018) for the proof.

The far right of the panel reports the mean return difference on the diagonal spread over positive E/P portfolios, a – b. As negative E/P cannot indicate a positive future earnings yield—the expected yield on a positive price there must be positive—the construction is limited to portfolios with positive E/P. The mean annual return spread is 13.3%. If the observed average returns are returns for risk, this indicates a significant risk difference between the long and short positions of a zero-net-investment factor formed on the accounting information.

We refer to this summary accounting factor as *SUMIS* as it is based on earnings from the income statement. Reservations about the crude estimate of expensed investment in SG&A prompted the same portfolio sort but with that component excluded from expensed investment, that is, with the $ExpInv_t/P_t$ (w/o SG&A) measure. Results are similar, with a mean return spread of 15.7% for the summary factor.[18]

4.2 SUMBS: A Balance Sheet Factor Based on Investment and Financing

SUMIS involves the accounting for current earnings that add to lagged booked value. But there is also a concurrent, time t, accounting in the balance sheet that, the accounting for investment activities and financing activities. As explained in Section 3.1, balance-sheet investment is the realization of uncertain investment opportunities that projects future realized earnings in

[18] Conservative accounting for investment does not affect earnings if the amount expensed is equal to the amortization charge that would be made if the investment were capitalized to the balance sheet and then amortized. This steady-state case occurs with no growth in the investment being expensed. So portfolios were also formed with the expensing variable in the third sort as $(ExpInv_t - ExpInv_{t-1})/P_t$, with little difference in results. Portfolios were also formed with the expensing variable as $\Delta CapExpInv_t/P_t$ which equals zero when expensed investment equals amortization, Again, results were similar. Table 5 indicated that the cash flow profitability measure, CP_t/P_t, performed relatively well, as did a CP_t/P_t factor, suggesting it might be a better indicator of the future earnings yield than E/P. So, the portfolio formation was repeated with CP_t/P_t, replacing E/P in the first sort. The return over the diagonal spread, a – b, was 17.4% (t = 2.24), higher than with E/P. However, this is just a finding in the data so, as we are unclear as to what CP_t/P_t captures (in the discussion of Table 5), we stick with the E/P sort. In any case, the finding does suggest incorporating a measure that conveys the expected yield better than E/P, a measure of "sustainable earnings."

equation (2), distinguished from more risky investment expensed against earnings. As a realization, it conveys lower risk and return, and the market prices it as such as Table 2 confirms. However, leverage with debt versus debt financing of investment adds to risk to future earnings in equation (2) and that leverage is also reported in the balance sheet. As double-entry accounting produces both a balance sheet and an income statement, the accounting for risk is not complete without this accounting in the balance sheet. This section forms a factor, SUMBS, that recognizes this accounting.

Most factor models such as Fama and French (2015 and 2018) and Hou, Xue, and Zhang (2015) measure investment as the change in total assets. However, as recognized in Cooper, Galen, and Ion (2024), investment so defined is not the investment in business assets such as inventory and plant. Indeed, while including that investment change in total assets also involves the change in cash and cash investments reduce cash with zero effect on total assets. So, Table 7 forms the balance-sheet factor based on investment in the net operating assets (in the business), ΔNOA.

Each March 31, firms are formed into three portfolios based on $\Delta NOA_t/P_{t-1}$ then within each $\Delta NOA_t/P_{t-1}$ portfolio, on financing leverage, $Debt_t/P_t$. $\Delta NOA_t/P_{t-1}$ ranks forward returns negatively in Table 7 while leverage is positively (though not strongly) related to returns. The mean return over years for SUMBS, calculated from the a – b diagonal, is 0.099 with a t-statistic of 3.06.[19]

4.3 A Size Factor?

[19] Similar results were observed with the investment calculated as (Change in gross property, plant, and equipment and inventory)$_t/P_{t-1}$ (yielding SUMBS = 0.072, t = 2.65). With investment as $\Delta TA_t/P_{t-1}$, SUMBS = 0.093, t = 2.21.

Both the Fama and French and *q*-investment models include a factor based on firm size (equity market capitalization) but without explanation. Should that factor be added to SUMIS and SUMBS? The answer would be No if size is picking up the risk conveyed by those numbers. Section 3.5 indicates that it does.

Table 8 investigates further. Panel A reports the median ln(Size) of firms in the portfolios underlying SUMIS in Table 6, showing that firm size varies over the accounting features underlying those portfolios. There is some negative correlation between size and E/P (across rows), but the significant differences are down columns: Small firms, prominent in row 3, are associated with lower realized ROE and higher expensed investment than large firms, on average. This accords with the mean correlations in Table 1 where the rank correlations of size with ROE is 0.41 and that with $ExpInv_t/P_t$ is -0.41.[20] This makes sense: Small firms typically are investing intensively in R&D, advertising, and other intangibles relative to the (low) earnings they are currently realizing, and thus are higher risk, while large firms are realizing earnings from those investments, thus lower risk. Indeed, large firms are typically once-small firms that became successful with realized earnings, whereas small firms are those where those investments have not yet paid off (and may not). Think again of the start-up biotech versus the mature pharmaceutical.

So, Panel B of Table 8 excludes microcaps (below the 20th size percentile of NYSE stocks) from the analysis in Panel B of Table 6 that yields the return to SUMIS. Results are not as strong, emphasizing that the excluded small firms are those with the features that convey higher risk. These findings indicate that the size premium in returns is partially explained by exposure

[20] The correlations with expensed investment in Table 1 are for $ExpInv_t/P_{t-1}$. Those reported in this section are for $ExpInv_t/P_t$.

to risk conveyed by the accounting; it is not an opaque size factor than explains returns but rather that exposure.

However, while the risk conveyed by the accounting is more associated with small firms, it is also associated with large firms with the same accounting characteristics. The portfolios in Panel C of Table 8 are the same as those in Panel B but the returns in the portfolios formed on ROE and $ExpInv_t/P_t$ (down columns) are only for firms that report the accounting measures that meet the cut-offs for those portfolios. The high minus low returns in the 3 – 1 row are significant as is the diagonal mean return spread of 0.091. It is not just a matter of the accounting measures explaining returns for small versus large firms but for any firm, small or large, with the relevant accounting numbers; size is a crude indicator of the differing risk between small and large firms.

These findings, along with those for Table 4 characteristic regressions partitioned on size, indicate that firm size proxies for risk conveyed by SUMIS. Comparative factor tests below further confirm. We have not documented the extent to which size returns are explained by SUMBS but that also is observed below.

4.4 A Tentative Factor Model

Adding the income statement and balance sheet factors to the excess return on the market portfolio, MKT, yields the following factor model:

$$R_{it} - R_t^{RF} = \alpha_{it} + \beta_{MKT,i} MKT_t + \beta_{SUMIS,i} SUMIS_t + \beta_{SUMBS,i} SUMBS_t + \varepsilon_{it}. \qquad (4)$$

where $R_{it} - R_t^{RF}$ is the excess return over the risk-free rate for asset i for period t. The market factor appears in most extant factor models with sensitivity to that portfolio the first-order measure of risk. The added accounting factors represent extra-market risk or capture part of

market risk if the proxy market portfolio does not represent the market portfolio of all assets adequately.

5. A Comparison of Accounting Factors with Extant Factor Models

We now compare this model with standard factor models for explaining returns. If (4) is a valid model, $\alpha_{it} = 0$, all i and t, the focus of the tests.

5.1 Standard Factor Returns Do Not Explain the Accounting Factor Returns

Table 9 reports results from time-series regressions of monthly excess returns for SUMIS and its underlying portfolios on factors in the Fama and French six-factor model. Panel A is for the high-risk portfolios within each E/P portfolio, those with low ROE with high expensed investment on row numbered 3 in Panel B of Table 6. Panel B is for the relatively low-risk portfolios, high ROE with low expensed investment on row numbered 1. Panel C is for the long minus short returns, row 3 minus 1. In all panels in Table 9, there is a portfolio consisting of all positive E/P stocks in the respective rows, labelled ALL. In the far-right column in Panels A and B, the return is the difference between the return on the high positive E/P and low positive E/P in the row. In Panel C, it is the return on SUMIS.

In these spanning tests, one might interpret the results as FF models explaining the returns for the accounting portfolios. Indeed, they do (in part) as a matter of correlation. But, with these models identified from correlations in the data, with the underlying risk they capture unexplained, that is not very illuminating. Rather, we take the analysis to ask if those factors capture the (explained) risk and returns conveyed by the accounting factors.

For the high and low risk portfolios in Panels A and B, the FF factors explain a good part of the variation in returns, with R^2 around 85%. We leave the reader to examine the slope coefficients

for the explanatory power of each FF factor, except to note that β_{SMB} on the SMB size factor is close to 1.0 for the high-risk portfolios in Panel A, indicating that the risk conveyed by size is similar to that from ROE and expensed investment (though β_{SMB} is conditional on other factors in the regression). The coefficients are lower for lower-risk portfolios in Panel B. They remain positive in Panel C, with SMB positively correlated with the SUMIS return. This attributes the size premium to the risk associated with the three accounting measures, complementing the size analysis in Section 3.5.

The Intercepts (alphas)—returns unexplained by the factors—are significantly different from zero, more so for the high-risk portfolios in Panel A. The alphas are positive, typically about 1% per month, with the return for the summary portfolio with zero net investment (in the last column) also 1% per month on average with a t-statistic of 7.73. This indicates that the FF factors do not fully capture the risk exposure for E/P yields with high-risk (that is, low ROE and high expensed investment). That is also the case in the lower-risk portfolios in Panel B, though the intercepts are not as high, reflecting the lower risk.

In Panel C where the spread on ROE and expensed investment is introduced, the alphas are also positive with the R^2 much reduced. The alpha for SUMIS is 0.7% per month (t = 4.42).[21]

[21] Two issues arise in these regressions. First, the FF factors are constructed excluding microcaps whereas the portfolios based on accounting measures include them. This is by design, for it is the accounting treatment to convey risk and return that we wish to document and, if this is associated more with small firms, so be it. Indeed, Panels A and B of Table 8 and the findings in Table 9 here so indicate. However, Panel C of Table 8 demonstrates that the risk is also in a large firms with the relevant accounting measures, requiring a higher return which an investor would want to identify. We repeated the analysis in Table 9 rejecting microcaps. Results were similar for slope coefficients, and significant intercepts were estimated. However, those intercepts were lower, 0.003 and 0.004 for the long minus short portfolios in Panel C and 0.003 with a t-statistic of 2.24 for SUMIS. Second, FF portfolio returns are value weighted whereas those constructed with accounting measures are equally weighted. If the accounting in focus is more associated with smaller firms, value weighting weights away from the accounting feature on which the portfolio was formed so the portfolio return is not indicative of the risk and return associated with that feature. Indeed, value weighting can be viewed as protecting against some of the risk indicated by accounting numbers and associated with smaller firms, by investing in larger firms where, on average, that risk is lower. That said, the alphas in Table 9 show that the accounting still explains returns in these value-weighted FF factors.

Table 10 repeats the same analysis but with the *q*-investment factors. Beside the market factor, those models have an investment factor, IA, a size factor, ME, an ROE factor, and, in a recent innovation in Hou, Mo, Xue, and Zhang (2021), a factor based on predicted investment growth in the future that the authors call EG. Again, the intercepts are significantly different from zero and about 1% per month for the higher-risk portfolios in Panel A. The alpha for SUMIS, in Panel C is 0.7% per month (t = 3.19).

We performed the same tests in Tables 9 and 10 with the balance-sheet factor, SUMBS. As both FF and *q*-investment models include an investment factor, albeit measured differently, we expected these models to perform relatively well in explaining return for SUMBS, but that is not so. For the FF factors, the alphas (and t-statistics) for the long and short components of SUMBS (corresponding to those in Panel C of Table 9) are 0.8% (6.84) and 0.3% (4.11) per month, respectively, and that for the long minus SUMBS factor itself is 0.5% (4.37). The corresponding numbers for the *q*-investment model are 0.9% (6.38), 0.4% (4.85), and 0.5% (3.51).

5.2 The Accounting Factor Model Explains Returns on a Variety of Test Portfolios

Table 11 reports on time-series regressions like those in Tables 9 and 10 but with monthly excess returns for selected portfolios now regressed on factors returns for MKT, SUMIS, and SUMBS in model (5). The Panel A portfolios are long minus short portfolios constructed as potential factors earlier in this paper. The portfolios in Panel B are factors in standard factor models.

The intercepts are of primary interest. Alphas are not significantly different from zero for B/P, E/P, Size, and ROE (long – short portfolios) in Panel A, and that for TotB/P (that includes expensed investment) only 0.002 per month. Nor are alphas significant for HML and SMB in

Panel B. The results for the Size and SMB are particularly compelling: The zero alpha indicates that the size premium reflects risk conveyed by the accounting factors, as indicated in the discussion in Section 4.3. The exceptions that stand out in Table 11 are the FF operating profitability factor, RMW, and the two factors from the *q* models, *q*-ROE and *q*-EG. Here alphas are relatively large, positive, and statistically significant.

The *q*-EG factor return is interpretable under accounting principles, indeed it validates them. In Hou, Mo, Xue, and Zhang (2021), this factor is calculated as the difference in returns between high forecasted growth in investment and low forecasted growth. Under Cochrane (1991) *q*-theory, higher expected investment growth (EG) requires a higher return as it indicates higher marginal *q* which equals marginal cost indicating the discount rate. Interpreted under accounting principles, future investment is investment not yet looked to the balance sheet, investment not yet realized, still at risk. That requires a higher return over that indicated by SUMIS and realized investment in SUMBS. Indeed, it complements SUMBS: While SUMBS books investment opportunities realized, EG captures investment growth opportunities still at risk.

So, both production-investment theory and accounting principles have the same prediction. Indeed, one can construe the accounting as consistent with production-investment theory. In any case, the finding that *q*-EG factor return is not explained by the accounting factor model (5) but is explained by accounting principles leads to an expected investment growth factor, EG, added to model (5):

$$R_{it} - R_t^{RF} = \alpha_{it} + \beta_{MKT,i}MKT_t + \beta_{SUMIS,i}SUMIS_t + \beta_{SUMBS,i}SUMBS_t + \beta_{EG,i}EG_t + \varepsilon_{it}. \quad (5)$$

As EG is predicted from time t accounting numbers in Hou, Mo, Xue, and Zhang (2021), it can be nominated as an accounting factor. Investment begets earnings so, with equation (2) defining

the expected return, EG captures expected future earnings growth from the expected investment, earnings still at risk for that investment is still at risk. That redefines the *q*-EG factor.[22]

Panels C and D of Table 11 repeats the analysis in Panels A and B of the table with the added EG factor return. β_{EG} is significant and positive, except for the Size and SMB (size) portfolios. Intercept alphas are close to zero for all portfolios, but positive for the size portfolios. That coincides with a negative β_{EG}: Conditional on the other factors, the relative returns for small firms are negatively correlated with the risk in anticipated investment.

6. Conclusions

This paper is offered with the aim of putting empirical asset pricing and the accounting on which it draws on the same platform. That is done by recognizing that accounting numbers are governed by accounting principles that connect the numbers to risk and expected return as a matter of pricing theory. This is so for book value, earnings, investment, return on equity (ROE), and other numbers that are the basis for factor construction in existing asset pricing models. In many of those models, these numbers enter from data mining without much definition. The paper provides that definition. But, in doing so, it recognizes the property of the double entry accounting system under which these numbers are codetermined and convey information about risk jointly. That leads to a parsimonious packaging of the accounting numbers into a factor model that differs from the standard factor model construction and outperforms these models in explaining returns.

That packaging begins with the recognition that, in theory, the expected return is given by the expected future earnings yield on current price, for which the current earnings-to-price, E/P, is a predictor. However, it is an imperfect predictor, for current earnings are typically disturbed by

[22] We concede that the revised model (5) comes from rationalization after observing the data, not a priori. Nevertheless, EG is rationalized as an implication of accounting principles.

accounting principles that convey the risk to expected earnings that discounts their pricing. By recognizing this accounting and combining it with the current earnings-to-price, the paper produces a factor that explains stock returns well relative to existing models. That is complemented with an investment factor based on complementary accounting in the balance sheet under double entry. Accounting measures in standard models enter, but in a different way, and in a way that conveys risk and return under accounting principles.

Here are some specific observations arising from the analysis and the empirical tests:

- The current earnings yield, E/P, is important for indicating risk and return but does not explicitly appear in standards models.
- As ROE and B/P appear in Fama and French models and E/P = ROE x B/P (approximately), those models capture E/P implicitly.
- Consequently, book value enters into risk models, not as a stand-alone B/P factor but as the denominator of ROE, adding information about risk and return to E/P.
- While those models and *q*-investment models incorporate ROE with a positive relationship to returns, the accounting for ROE implies a negative relationship to returns.
- The determining accounting is the realization principle and the expensing of particularly risky investment that reduces earnings and ROE (and other profitability measures). That leads to an accounting factor based on E/P, ROE, and expensed investment based on the income statement.
- Investment booked to the balance sheet, deemed less risky, is the realization of investment opportunities and thus conveys lower risk. That supplies an accounting justification for the investment factor in extant factor models, though requires a different measure of investment to change in total assets.

- The construction of factors that adds expensed investment to balance-sheet investment loses information about the differential risk in these two types of investment. So do factors that add book value from expensed investment to B/P.
- That balance-sheet investment combines with financing leverage to yield a balance-sheet factor additive to the income statement factor under double-entry accounting.
- Expected investment, unrealized and still at risk under accounting principles, indicates higher risk. This supplies an accounting justification for the expected investment growth factor in *q*-investment models.

The paper does not provide a definitive factor model. With the aim of making comparisons to existing models it has been limited to accounting measures appearing in those models. Accounting principles affect a wider set of accounting items that could be exploited to build a model. Those items add to the prediction of the future earnings yield over that in the current earnings in E/P. However, determined under accounting principles, they also convey the risk to that growth. Penman and Zhu (2022) pursues this, not only generating a factor model, but also casting it in terms of consumption-based asset pricing theory with accounting numbers connecting to consumption and the risk to consumption.

Appendix. Calculation of Variables

This appendix details the calculation of variables used in the empirical analysis. Compustat sometimes reports an item as missing. These are either in the financial reports as zero, aggregated with other items in the financial reports, or reported but not found by Compustat. We follow the practice of setting these to zero, relying on the finding in Casey, Gao, Kirschenheiter, Li, and Pandit (2016) that setting to zero satisfies certain articulating accounting equations 82.7% of the time. Nevertheless, the remaining 17.3% are a qualification to our analysis.

Ret_{t+1}	One-year, buy-and-hold return calculated as compounded CRSP monthly returns, starting at the beginning of the fourth month after the current fiscal year end. For firms that are delisted during the 12 months, the return for the remaining months is calculated by first applying the CRSP delisting return and then reinvesting any remaining proceeds at the risk-free rate. This mitigates concerns with potential survivorship biases. Firms that are delisted for poor performance (delisting codes 500 and 520-584) frequently have missing delisting returns (see Shumway 1997). We control for this potential bias by applying delisting returns of -100% in such cases. Results are qualitatively similar if we make no such adjustment. Accounting data for a fiscal year are presumed to have been published during the three months after fiscal-year end (and before the return period), as required by law (and before the beginning of the return period).
$Earnings_t$	Earnings before extraordinary items (Compustat item IB) and special items (Compustat item SPI), minus preferred dividends (Compustat item DVP), with a tax allocation to special items at the prevailing federal statutory corporate income tax rate for the year. The top statutory federal tax rate was 50% in 1964, 48% in 1965-1967, 52.8% in 1968-1969, 49.2% in 1970, 48% in 1971-1978, 46% in 1979-1986, 40% in 1987, 34% in 1988-1992, 35% in 1993-2017, and 21% in 2018.
B_t	Book value of common equity at the end of the fiscal-year, calculated as Compustat common equity (Compustat item CEQ) plus any preferred treasury stock (Compustat item TSTKP) less any preferred dividends in arrears (Compustat item DVPA).
P_t	Market value of equity three months after fiscal-year end. It is calculated as the number of shares outstanding multiplied by the price per share from CRSP at three months after fiscal-year end. It can also be calculated as the number of shares outstanding at the end of the fiscal year from Compustat (#CSHO) multiplied by the price per share from CRSP at three months after fiscal-year end, adjusted for any intervening stock splits and stock dividends. This excludes any change in the market price from net share issues over the three months when calculating E/P and B/P ratios based on earnings and book value at year end. Market price can also be calculated as per-share price at three months after fiscal-year end, adjusted for stock splits and stock dividends over the three months, in which case, earnings and book value are on a per-share basis. The first calculation is applied in this paper as that seems to be the practice for E/P and B/P calculations in asset pricing papers to which this paper compares.
TA_t	Total assets (Compustat item AT).

$Investment_t$	Change in Total Assets. This is the measure in Fama and French (2015 and 2018) five- and six-factor models and the Hou, Xue, and Zhang (2015) and Hou, Mo, Xue, and Zhang (2021).
$ExpInv_t$	$ExpInv_t = R\&D_t + Advertising_t + SG\&A_t$ x 0.3. R&D is Compustat item XRD, Advertising is Compustat item XAD and the SG&A variable is XSGA-XAD-XRD when XSGA is greater than the sum of XAD and XRD, otherwise SG&A equals XSGA. The modified measure, $ExpInv_t$ (w/o SG&A) excludes the SG&A component.
$CapExpInv_t$	Capitalized expensed investment, calculated as the sum of capitalized R&D, capitalized advertising expense, and capitalized SG&A x 0.3. R&D is capitalized with amortization rates estimated in Lev and Sougiannis (1996). Advertising expenses are capitalized using the sum-of-years digit amortization method with a two-year useful life, as indicated by the findings in Bublitz and Ettredge (1989) and Hall, Mansfield, and Jaffe (1993). SG&A is capitalized and amortized using the straight-line method with a five-year useful life.
$TotB_t$	Book value plus capitalized expensed investment, $B_t + CapExpInv_t$.
$Debt_t$	Financing debt at the end of the fiscal year calculated as Debt in current liabilities (#DLC) + Long-term debt (#DLTT) + Preferred stock (#PSTK) – Preferred treasury stock (#TSTKP) + Preferred dividends in arrears (#DVPA).
Net $Debt_t$	Net financing debt at the end of the fiscal year, the difference between financing debt (above) and financial assets. Financial assets are Cash and Short-term investments (#CHE) + Investments and advances-other (#IVAO).
NOA_t	Net operating assets at the end of the fiscal year, measured as Net $Debt_t$ (above) plus book value of common equity, B_t, with added non-controlling (minority) interest (Compustat item MIB).
ROE_t	$Earnings_t/B_{t-1}$, with $Earnings_t$ calculated as above.
ROE_t^{Adj}	$\frac{(\text{Earnings} + \text{Expensed Investment} - \text{Amortized CapExpInv})_t}{TotB_{t-1}}$
GP_t	Gross profit, as in Novy-Marx (2013) where it is relative to total assets.
CP_t	Cash flow profit as in Ball, Gerakos, Linnainmaa, and Nikolaev (2018) where it is relative to lagged total assets. In Fama and French (2018) it is deflated by end-of-period book value.
OP_t	Operating profits as in Fama and French RMW factor. That is defined as revenue minus cost of goods sold minus SG&A minus interest on the Kenneth French web site, though it appears to include R&D in Fama and French (2018). On the web site OP_t is deflated by lagged book value, but in Fama and French (2018) it is deflated by end-of-period book value for the construction of RMW.
$Beta_t$	Stock market betas estimated from monthly stock returns during the previous five years ending in the third month after the current fiscal year end, or a lessor period if five years of data are not available.
$Size_t$	Logarithm of market value of $equity_t$.
MOM_t	Twelve month buy-and-hold return ending in the third month after the current fiscal year end.

References

Arnott, R., C. Harvey, V. Kalesnik, and J. Linnainmaa. (2021). Reports of value's death may be greatly exaggerated. *Financial Analysts Journal* 77 (1), 44-67.

Ball, R. (1978). Anomalies in relationships between securities' yields and yield-surrogates. *Journal of Financial Economics* 6, 103-126.

Ball, R., J. Gerakos, J. Linnainmaa, and V. Nikolaev. (2015). Deflating profitability. *Journal of Financial Economics* 117, 225-248.

Ball, R., J. Gerakos, J. Linnainmaa, and V. Nikolaev. (2018). Profitability, cash flows and accruals in the cross section of stock returns. *Journal of Financial Economics* 121, 28-45.

Ball, R., J. Gerakos, J. Linnainmaa, and V. Nikolaev. (2020). Earnings, retained earnings, and book-to-market in the cross-section of stock returns. *Journal of Financial Economics* 135, 231-254.

Bansal, R., and A. Yaron. (2004). Risks for the Long-run: A potential resolution of asset pricing puzzles. *Journal of Finance* 59, 1481-1509.

Barker, R., A. Lennard, S. Penman, and A. Teixeira. (2021). Accounting for intangible assets: Suggested solutions. *Accounting and Business Research*, forthcoming.

Basu, S. (1977). Investment performance of common stocks in relation to their price-earnings ratios: A test of the efficient market hypothesis. *The Journal of Finance* 32, 663-682.

Basu, S. (1983). The relationship between earnings yield, market value, and return for NYSE stocks: further evidence. *Journal of Financial Economics* 12, 129-56.

Berk, J., R. Green, and V. Naik. (1999). Optimal investment, growth options, and security returns. *Journal of Finance* 54, 1153-1608.

Bongaerts, D., X. Kang, and M. van Dijk. (2021). Intangible assets and the cross-section of returns. Unpublished paper, Rotterdam School of Management, Erasmus University. At https://ssrn.com/abstract=3927990.

Bublitz, B., and M. Ettredge. (1989). The information in discretionary outlays: Advertising, research, and development. *The Accounting Review* 64 (1):108-124.

Casey, R. J., F. Gao, M. T. Kirschenheiter, S. Li, and S. Pandit. (2016). Do Compustat financial statement data articulate? *Journal of Financial Reporting* 1 (1): 37-59.

Cho, T., L. Kremens, D. Lee, and C. Polk. (2021). Scale or yield? A present-value identity. Unpublished paper, London School,of Economics, University of Washington, and UCLA Anderson. At https://ssrn.com/abstract=3857368.

Cochrane, J. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance* 46¸ 209-237.

Cooper, M., H. Gulen, and M. Schill. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance* 63, 1609-1651.

Cooper, M., H. Gulen, and M. Ion. (2024). The use of asset growth in empirical asset pricing models. *Journal of Financial Economics* 151, 103746.

Duan, Z., Z. Gong, and Q. Qi. (2021). Factorization asset pricing. Unpublished paper, Peking University. At https://ssrn.com/abstract=3940074.

Eisfeldt, A., and D. Papanikolaou. (2013). Organizational capital and the cross-section of expected returns. *Journal of Finance* 67 (4), 1365-1406.

Eisfeldt, A., E. Kim, and D. Papanikolaou. (2020). Intangible value. Unpublished paper at https://ssrn.com/abstract=3720983.

Epstein, L., and S. Zin. (1989). Substitution, Risk aversion, and the temporal behavior of consumption and asset returns. *Econometrica* 57, 937-969.

Dechow, P., R. Sloan, and M. Soliman. (2004). Implied equity duration: A new measure of equity risk. *Review of Accounting Studies* 9, 197-228.

Fama, E. and K. French. (1992). The cross-section of expected returns. *Journal of Finance* 47 (2), 427-465.

Fama, E., and K. French. (1993). Common risk factors in the returns of stocks and bonds. *Journal of Financial Economics* 33, 3-56.

Fama, E., and K. French. (2006). Profitability, investment and average returns. *Journal of Financial Economics* 82, 491-518.

Fama, E., and K. French. (2015). A five-factor asset pricing model. *Journal of Financial Economics* 116, 1-22.

Fama, E., and K. French. (2018). Choosing factors. *Journal of Financial Economics* 128, 234-252.

Fama, E., and K. French. (2020). Comparing cross-section and time-series factor models. *Review of Financial Studies* 33, 1891-1926.

Feltham, G. and J. Ohlson. (1995). Valuation and clean surplus accounting for operating and financial activities. *Contemporary Accounting Research* 11, 689-731.

Hall, B. H., E. Mansfield, and A. B. Jaffe. (1993). Industrial research during the 1980s: Did the rate of return fall? *Brookings Papers on Economic Activity. Microeconomics* 1993 (2):289-343.

Hou, K., H. Mo, C. Xue, and L. Zhang. (2021). An augmented *q*-factor model with expected growth. *Review of Finance* 25 (1), 1-41.

Hou, K., C. Xue, and L. Zhang. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies* 28, 650-705.

Kapons, M. (2020). R&D expenses, R&D capitalization, the book-to-price ratio and the cross-section of returns. Working paper, Tilburg School of Economics and Management, Tilburg University. Presented at the Accounting Design Project Forum. At https://www8.gsb.columbia.edu/ceasa/sites/ceasa/files/20200918_ADP_Kapons.pdf.

Lev, B., and Sougiannis, T. (1996). 'The capitalization, amortization, and value relevance of R&D'. *Journal of Accounting and Economics*, *2* (February): 107-138.

Lev, B., and A. Srivastava. (2020). Explaining the recent failure of value investing. Working paper, New York University and University of Calgary. At https://ssrn.com/abstract=3442539.

Li, F. (2020). Intangibles: The missing ingredient in book value. Paper, Research Affiliates, LLC. At https://ssrn.com/abstract=3686595.

Merton, R. (1973). An intertemporal asset pricing model. *Econometrica* 41, 867-887.

Mullins, G. (2020). Equity duration. Unpublished paper, St. Cross College, University of Oxford.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics* 108, 1-28.

Oh, H., and S. Penman. (2024). Income statement mismatching has not reduced the informativeness of earnings over time. *Journal of Business Finance and Accounting* 51, 756-782.

Ohlson, J. (1995). Earnings, book values, and dividends in equity valuation. *Contemporary Accounting Research* 12, 661-687.

Oswald, D., A. Simpson, and P. Zarowin. (2020). The information benefits of R&D capitalization. Woking paper, University of Michigan, London School of Economics, and New York University. At https://ssrn.com/abstract=

Park, H. (2022). An intangible-adjusted book-to-market ratio still predicts stock returns, *Critical Finance Review* 11,265-297.

Penman, S. (2021). Accounting for Risk. *Foundations and Trends in Accounting* 16 (1-2), 1-135.

Penman, S. (2025). Empirical research on capitalizing intangible assets is logically incoherent. At https://ssrn.com/abstract=4982366.

Penman, S., and F. Reggiani. (2018). Fundamentals of value vs. growth investing and an explanation for the value trap. *Financial Analysts Journal* 74, 102-119.

Penman, S., and F. Reggiani. (2024). A fundamental explanation for the size effect in returns and its variation over time, At https://ssrn.com/abstract=4180896.

Penman, S., F. Reggiani, S. Richardson, and İ Tuna. (2018). A framework for identifying accounting characteristics for asset pricing models, with an evaluation of book-to-price. *European Financial Management* 24, 488-520.

Penman, S., and N. Yehuda. (2019). A matter of principle: The identification of cash-flow news and discount-rate news in financial statements. *Management Science* 65, 5584-5602.

Penman, S. and X. Zhang. (2020). A theoretical analysis connecting conservative accounting to the cost of capital. *Journal of Accounting and Economics* 69, 1-25.

Penman, S. and X. Zhang. (2021). Connecting book rate of return to risk and return: The information conveyed by conservative accounting. *Review of Accounting Studies* 26 (1), 391-423.

Penman, S. and X. Zhang. (2025). Deriving Fama and French factors. At https://ssrn.com/abstract=4895017.

Penman, S., and J. Zhu. (2022). An accounting-based asset pricing model and a fundamental factor. *Journal of Accounting and Economics* 73 (2-3), article 101476.

Peters, R., and L. Taylor. (2017). Intangible capital and the investment-*q* relation. *Journal of Financial Economics* 123, 251-272.

Rizova, S., and N. Saito. (2020). Intangibles and expected stock returns. Unpublished paper, Dimensional Fund Advisors.

Shumway, T. (1997). The delisting bias in CRSP data. *Journal of Finance* 52, 327-340.

Titman, S., K. Wei, and F. Xie. (2004). Capital investment and stock returns. *Journal of Financial and Quantitative Analysis* 39, 677-700.

Weber, M. (2018). Cash flow duration and the term structure of equity returns. *Journal of Financial Economics* 128 (3), 486-503.

Zhang, X. (2000). Conservative accounting and equity valuation. *Journal of Accounting and Economics* 29, 125-149.

Table 1: Distribution of Variables and Correlation Matrix

Panel A summarizes the average of annual cross-sectional distributions from 1964 to 2018 for variables appearing in the paper. For the calculation of means and standard deviations, variables are trimmed at the top and bottom one percent, each year, except for returns. N is the average number of observations per year. For ROE_t, firms with negative lagged book values are omitted. Panel B reports means of annual cross-sectional correlation coefficients across the sample years. Pearson correlations are presented in the upper off-diagonal and Spearman correlations in the lower off-diagonal. For the Pearson correlations, all variables are trimmed at the top and bottom 1 percent, except for returns. R^{RF} is the risk-free rate. Variables are defined in the appendix.

Panel A: Distributions of Variables

Variable	N	Mean	Median	Std. Dev.	P5	P25	P75	P95
$Investment_t/TA_{t-1}$	3,163	0.213	0.078	0.579	-0.233	-0.019	0.227	1.120
$Investment_t/P_{t-1}$	2,907	0.083	0.075	0.499	-0.566	-0.032	0.208	0.759
$ExpInv_t/TA_{t-1}$	3,078	0.159	0.111	0.168	0.012	0.054	0.203	0.474
$ExpInv_t$ (w/o SG&A)$/TA_{t-1}$	2,345	0.091	0.043	0.138	0.000	0.013	0.108	0.349
$ExpInv_t/P_{t-1}$	2,830	0.202	0.123	0.246	0.014	0.059	0.243	0.669
B_t/P_t	3,106	0.794	0.618	0.688	0.129	0.356	1.004	2.059
$CapExpInv_t/P_t$	3,054	0.316	0.139	0.621	0.014	0.062	0.304	1.153
$CapExpInv_t$ (w/o SG&A)$/P_t$	2,494	0.089	0.034	0.159	0.000	0.010	0.094	0.367
$TotB_t/P_t$	3,106	1.120	0.804	1.159	0.194	0.472	1.332	3.089
ROE_t	3,035	-0.020	0.090	0.536	-0.778	-0.058	0.178	0.408
OP_t/B_{t-1}	3,035	0.200	0.237	0.563	-0.575	0.062	0.394	0.845
CP_t/TA_{t-1}	2,900	0.109	0.131	0.211	-0.234	0.048	0.210	0.377
E_t/P_t	3,216	-0.058	0.049	0.402	-0.603	-0.036	0.085	0.156
$\Delta B_t/B_{t-1}$	3,035	0.251	0.079	0.992	-0.439	-0.037	0.213	1.405
$CapExpInv_t/NOA_t$	2,943	1.117	0.357	3.112	0.035	0.163	0.764	3.964
$Beta_t$	3,017	1.242	1.161	0.726	0.191	0.769	1.639	2.578
$Size_t$	3,216	4.632	4.504	1.926	1.665	3.227	5.906	8.074
$Momentum_t$ (MOM_t)	3,026	0.165	0.053	0.685	-0.543	-0.204	0.364	1.216
Ret_{t+1}	3,216	0.161	0.056	0.696	-0.575	-0.210	0.368	1.213

Panel B: Average Pearson Correlations (above the diagonal) and Spearman Correlations (below the diagonal) between Variables

	$Investment_t/P_{t-1}$	$ExpInv_t/P_{t-1}$	B_t/P_t	E_t/Pt	$CapExpInv_t/P_t$	ROE_t	OP_t/B_{t-1}	CP_t/TA_{t-1}	$Beta_t$	$Size_t$	MOM_t	$Ret_{t+1}-R^{RF}_{t+1}$
$Investment_t/P_{t-1}$		-0.083	-0.075	0.371	-0.171	0.201	0.188	-0.113	-0.015	0.121	0.144	-0.023
		(-4.09)	(-5.06)	(23.00)	(-10.35)	(14.84)	(17.47)	(-5.14)	(-1.44)	(9.10)	(12.80)	(-2.82)
$ExpInv_t/P_{t-1}$	-0.048		0.365	-0.172	0.615	-0.147	-0.106	-0.102	-0.015	-0.308	0.098	0.033
	(-2.50)		(41.75)	(-11.18)	(26.32)	(-12.67)	(-10.22)	(-8.13)	(-1.14)	(-33.22)	(6.88)	(3.32)
B_t/P_t	-0.055	0.333		-0.137	0.374	-0.130	-0.182	-0.139	-0.086	-0.331	-0.267	0.070
	(-4.21)	(26.01)		(-5.31)	(19.66)	(-4.54)	(-7.28)	(-7.94)	(-6.64)	(-32.23)	(-27.96)	(5.48)
E_t/P_t	0.348	-0.035	0.194		-0.328	0.480	0.346	0.225	-0.084	0.269	0.181	0.025
	(28.62)	(-1.45)	(7.82)		(-12.01)	(32.07)	(27.98)	(19.73)	(-7.29)	(15.91)	(10.21)	(2.07)
$CapExpInv_t/P_t$	-0.182	0.828	0.458	-0.083		-0.170	-0.159	-0.128	-0.002	-0.358	-0.188	0.044
	(-8.95)	(121.11)	(38.44)	(-3.05)		(-15.34)	(-15.97)	(-15.16)	(-0.16)	(-32.02)	(-15.66)	(3.79)
ROE_t	0.339	-0.231	-0.326	0.652	-0.348		0.825	0.481	-0.058	0.272	0.186	0.042
	(22.76)	(-15.20)	(-12.52)	(19.49)	(-30.77)		(144.30)	(33.04)	(-3.96)	(30.26)	(12.22)	(3.58)
OP_t/B_{t-1}	0.314	-0.183	-0.319	0.533	-0.308	0.858		0.536	-0.024	0.287	0.203	0.041
	(24.99)	(-11.72)	(-14.26)	(18.65)	(-26.85)	(273.45)		(32.88)	(-1.49)	(28.54)	(14.47)	(3.80)
CP_t/TA_{t-1}	-0.038	-0.157	-0.256	0.332	-0.216	0.527	0.584		-0.049	0.303	0.126	0.069
	(-1.48)	(-9.93)	(-16.03)	(16.03)	(-18.23)	(52.01)	(55.35)		(-4.18)	(32.58)	(9.59)	(5.39)
$Beta_t$	-0.00	0.006	-0.101	-0.138	0.003	-0.054	-0.046	-0.044		0.013	0.004	-0.038
	(-0.03)	(0.39)	(-6.44)	(-8.47)	(0.16)	(-2.75)	(-2.32)	(-3.55)		(0.69)	(0.21)	(-2.71)
$Size_t$	0.164	-0.344	-0.327	0.239	-0.463	0.407	0.411	0.352	0.038		0.171	-0.028
	(10.52)	(-23.24)	(-23.33)	(8.84)	(-39.91)	(32.72)	(29.06)	(38.06)	(1.85)		(9.63)	(-1.63)
MOM_t	0.213	0.074	-0.333	0.173	-0.289	0.325	0.298	0.211	-0.044	0.271		-0.008
	(21.71)	(4.36)	(-23.55)	(6.84)	(-17.10)	(20.68)	(19.70)	(16.53)	(-1.83)	(13.52)		(-0.68)
$Ret_{t+1} - R^{RF}_{t+1}$	-0.004	0.033	0.087	0.147	0.042	0.092	0.099	0.123	-0.065	0.059	0.039	
	(-0.40)	(2.26)	(5.59)	(9.45)	(2.79)	(5.72)	(6.41)	(9.01)	(-3.95)	(2.93)	(2.25)	

Table 2: Investment and Forward Stock Returns

Panel A reports mean coefficient estimates from annual cross-sectional decile-rank regressions of excess annual forward stock returns, Ret_{t+1} over the risk-free rate, on accounting numbers involving investment, 1964-2018. t-statistics (in parentheses) are calculated as the mean coefficients relative to their standard errors estimated from the time-series of 55 coefficient estimates. Panel B reports mean excess annual returns for portfolios formed from ranking on selected investment measures each March 31st, along with mean return differences and t-statistics based on standard errors calculated from the time series of return differences between the high and low portfolios. Variables are defined in the appendix. For regressions with ROE, firms with negative lagged book values are excluded.

Panel A: Decile Rank Regression Results

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.173	0.102	0.101	0.070	0.028	0.040
	(4.21)	(3.35)	(2.64)	(1.39)	(0.61)	(0.73)
$Investment_t/TA_{t-1}$	-0.013					
	(-4.51)					
$ExpInv_t/TA_{t-1}$		0.003				
		(0.78)				
$Investment_t/P_{t-1}$			-0.009	-0.011	-0.008	-0.004
			(-2.95)	(-5.50)	(-3.00)	(-1.93)
$ExpInv_t/P_{t-1}$			0.012	0.014	0.014	0.008
			(4.95)	(5.72)	(5.77)	(3.37)
ROE_t				0.008		
				(2.03)		
CP_t/TA_{t-1}					0.013	0.018
					(5.14)	(10.22)
$Beta_t$						0.000
						(0.06)
B_t/P_t						0.011
						(3.78)
$Size_t$						-0.010
						(-2.67)
MOM_t						-0.006
						(-1.22)
Adjusted R^2	0.007	0.008	0.014	0.024	0.024	0.060

Panel B: Mean Returns for "Factor-mimicking" Portfolios Formed from Ranking on Investment Measures

The High (Low) portfolio consists of firms in the top (bottom) 1/3 of the ranked number each year.

	Portfolios formed by ranking on			
	$\frac{Investment_t}{TA_{t-1}}$	$\frac{ExpInv_t}{TA_{t-1}}$	$\frac{Investment_t}{P_{t-1}}$	$\frac{ExpInv_t}{P_{t-1}}$
Low	0.158	0.103	0.154	0.067
Medium	0.124	0.117	0.114	0.121
High	0.065	0.128	0.090	0.171
High-Low	-0.093	0.025	-0.064	0.105
	(-4.16)	(0.86)	(-2.44)	(4.31)

Table 3: Book Value and Forward Stock Returns

Panel A reports mean coefficient estimates from annual cross-sectional decile-rank regressions of forward excess stock returns, Ret_{t+1} over the risk-free rate, on accounting numbers involving book value measures, 1964-2018. t-statistics (in parentheses) are calculated as the mean coefficients relative to their standard errors estimated from the time-series of 55 coefficient estimates. Panel B reports mean excess returns for portfolios from ranking on selected measures of book value-to-price each March 31st, along with mean return differences and t-statistics based on standard errors calculated from the time series of return differences between the high and low portfolios. Variables are defined in the appendix. For regressions with ROE, firms with negative lagged book values are excluded.

Panel A: Decile Rank Regression Results

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.033	0.023	-0.072	-0.096	0.005	-0.014
	(1.03)	(0.93)	(-1.69)	(-2.66)	(0.10)	(-0.26)
B_t/P_t	0.018		0.013	0.013	0.010	0.010
	(4.49)		(3.40)	(3.44)	(3.13)	(3.13)
$CapExpInv_t/P_t$		0.020	0.017	0.017	0.014	0.013
		(4.98)	(5.32)	(4.66)	(5.27)	(4.60)
ROE_t			0.011		0.013	
			(3.11)		(4.49)	
CP_t				0.016		0.017
				(6.92)		(9.36)
$Beta_t$					-0.000	0.000
					(-0.11)	(0.07)
$Size_t$					-0.008	-0.009
					(-2.03)	(-2.31)
MOM_t					-0.003	-0.002
					(-0.80)	(-0.57)
Adjusted R^2	0.015	0.013	0.033	0.036	0.056	0.058

Panel B: Mean Returns for Portfolios Formed from Ranking on Book Value Measures

The High (Low) portfolio consists firms in the top (bottom) 1/3 of the ranked number each year.

	Portfolios formed by ranking on		
	$\frac{B_t}{P_t}$	$\frac{CapExpIn_t}{P_t}$	$\frac{TotB_t}{P_t}$
Low	0.070	0.059	0.064
Medium	0.112	0.110	0.108
High	0.162	0.175	0.170
High-Low	0.091	0.116	0.106
	(3.17)	(3.66)	(3.48)

Table 4: ROE and Forward Stock Returns

The table reports mean coefficient estimates from annual cross-sectional decile-rank regressions of forward excess stock returns, Ret_{t+1} over the risk-free rate, on accounting numbers involving ROE and its components, 1964-2018. t-statistics (in parentheses) are calculated as the mean coefficients relative to their standard errors estimated from the time-series of 55 coefficient estimates. Panel A reports results for firms with positive ROE, Panel B for negative ROE. Panel C reports mean excess annual returns on portfolios formed each March 31on E_t/P_t, ROE_t, and $ExpInv_t/P_t$, along with the mean return differences (with t-statistics) between high and low portfolios. Variables are defined in the appendix. For regressions with ROE, firms with negative lagged book values are excluded.

Panel A: Decile Rank Regression Results for Positive Earnings Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.015	0.002	0.064	0.124	0.059	0.052	0.133	0.222	0.131
	(4.44)	(0.06)	(2.20)	(3.49)	(1.90)	(1.74)	(3.37)	(5.42)	(3.37)
ROE_t	-0.005	0.007		-0.011	-0.002	-0.004	-0.001	0.005	0.002
	(-2.26)	(2.47)		(-3.33)	(-0.92)	(-1.53)	(-0.37)	(2.52)	(0.80)
B_t/P_t		0.017			0.009				
		(4.47)			(2.81)				
E_t/P_t			0.012	0.014	0.009	0.010	0.007		0.007
			(4.38)	(4.48)	(3.31)	(3.19)	(2.54)		(2.22)
$ExpInv_t/P_t$						0.013	0.009		0.009
						(4.46)	(3.72)		(3.67)
$Investment_t/TA_{t-1}$								-0.012	-0.008
								(-6.04)	(-4.82)
$Beta_t$							-0.001		
							(-0.40)		
$Size_t$							-0.010	-0.013	-0.010
							(-3.04)	(-3.96)	(-2.97)
MOM_t							-0.002		
							(-0.65)		
Adjusted R^2	0.004	0.019	0.013	0.020	0.022	0.028	0.056	0.028	0.044

Panel B: Decile Rank Regression Results for Negative Earnings Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.076	0.009	0.144	0.143	0.098	-0.018	-0.020	0.143	0.001
	(1.32)	(0.16)	(2.18)	(2.06)	(1.37)	(-0.29)	(-0.27)	(2.13)	(0.02)
ROE_t	0.037	0.012		0.085	0.055	0.056	0.052	0.042	0.049
	(3.57)	(0.99)		(6.60)	(5.64)	(4.52)	(4.92)	(4.38)	(3.87)
B_t/P_t		0.019			0.009				
		(3.28)			(2.29)				
E_t/P_t			-0.008	-0.027	-0.020	-0.011	-0.010		-0.009
			(-1.25)	(-3.12)	(-2.69)	(-1.43)	(-1.75)		(-1.06)
$ExpInv_t/P_t$						0.022	0.025		0.022
						(4.71)	(4.89)		(4.93)
$Investment_t/$ TA_{t-1}								-0.011	-0.005
								(-3.83)	(-1.54)
$Beta_t$							-0.000		
							(-0.11)		
$Size_t$							-0.010	-0.015	-0.005
							(-1.07)	(-2.09)	(-0.85)
MOM_t							-0.003		
							(-0.46)		
Adjusted R^2	0.007	0.020	0.005	0.028	0.033	0.032	0.023	0.020	0.036

Panel C: Mean Returns for Portfolios formed on E_t/P_t, ROE_t, and $ExpInv_t/P_t$

The High (Low) portfolio consists of the top (bottom) 1/3 of the ranked number each year.

	Negative E/P	Positive E/P	ROE	$ExpInv_t/P_t$
Low	0.143	0.097	0.118	0.057
Medium	0.093	0.112	0.124	0.112
High	0.077	0.141	0.110	0.174
High - Low	-0.066	0.044	-0.008	0.117
	(-1.66)	(2.33)	(-0.24)	(3.53)

Table 5: Alternative Profitability Measures and Forward Stock Returns

Panels A and B report mean coefficient estimates from annual cross-sectional decile-rank regressions of forward excess stock returns, Ret_{t+1} over the risk-free rate, on accounting numbers that include operating profitability, OP_t/B_{t-1}, cash flow profitability, CP_t/TA_{t-1}, and a profitability measure, ROE_t^{Adj}, that corrects for the effect of conservative accounting on ROE, 1964-2018. Panel A is for firms with profits, Panel B for loss firms. t-statistics (in parentheses) are calculated as the mean coefficients relative to their standard errors estimated from the time-series of 55 coefficient estimates. Panel C reports mean excess annual returns for portfolios formed each March 31 on CP_t/P_t and OP_t/P_t, along with mean return differences (with t-statistics) between high and low portfolios. Variables are defined in the appendix. For regressions with OP_t/B_{t-1} and ROE_t, firms with negative lagged book values are excluded.

Panel A: Decile Rank Regression Results for Positive Earnings Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Intercept	0.128	0.079	0.155	0.121	0.097	0.132	0.050	0.008	0.059	0.132	0.095	0.126	-0.005	-0.028
	(3.98)	(2.27)	(4.62)	(3.32)	(2.54)	(3.65)	(1.65)	(0.23)	(1.96)	(3.33)	(2.28)	(3.38)	(-0.17)	(-0.83)
OP_t/B_{t-1}	-0.002			0.006			0.005			0.005				
	(-0.71)			(1.70)			(1.39)			(1.65)				
CP_t/TA_{t-1}		0.007			0.011			0.012			0.013	0.010	0.011	
		(3.06)			(4.46)			(4.95)			(5.55)	(3.62)	(4.67)	
CP_t/P_t														0.015
														(5.99)
ROE_t^{Adj}			-0.006			-0.006			-0.005					
			(-3.14)			(-2.58)			(-1.95)					
ROE_t				-0.016	-0.017	-0.007	-0.009	-0.009	-0.001	-0.005	-0.006	-0.010	-0.003	0.003
				(-3.35)	(-4.79)	(-1.86)	(-1.82)	(-3.02)	(-0.32)	(-1.26)	(-1.94)	(-3.88)	(-1.10)	(1.44)
E_t/P_t				0.014	0.015	0.014	0.010	0.010	0.010	0.006	0.006			
				(4.32)	(4.84)	(4.46)	(3.03)	(3.29)	(3.16)	(2.40)	(2.56)			
$ExpInv_t/P_t$							0.012	0.015	0.013	0.008	0.010		0.019	0.014
							(4.39)	(5.07)	(4.50)	(3.62)	(4.10)		(6.13)	(4.60)
$Beta_t$										-0.001	-0.001			
										(-0.36)	(-0.31)			
$Size_t$										-0.011	-0.012			
										(-3.17)	(-3.46)			
MOM_t										-0.003	-0.003			
										(-0.71)	(-0.73)			
Adjusted R^2	0.003	0.008	0.004	0.022	0.029	0.021	0.030	0.037	0.030	0.058	0.063	0.013	0.027	0.030

Panel B: Decile Rank Regression Results for Negative Earnings Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Intercept	0.079	0.057	0.080	0.132	0.099	0.136	-0.028	-0.054	-0.014	-0.125	-0.045	0.040	-0.120	-0.108
	(1.56)	(1.14)	(1.36)	(1.95)	(1.50)	(1.95)	(-0.47)	(-0.93)	(-0.21)	(-0.81)	(-0.66)	(0.70)	(-2.51)	(-2.30)
OP_t/B_{t-1}	0.020			0.016			0.016			0.010				
	(3.43)			(2.94)			(2.82)			(1.23)				
CP_t/TA_{t-1}		0.021			0.018			0.014			0.012	0.017	0.013	
		(5.50)			(4.50)			(3.43)			(2.13)	(4.12)	(2.93)	
CP_t/P_t														0.009
														(3.12)
ROE_t^{Adj}			0.025			0.041			0.013					
			(1.18)			(2.07)			(0.55)					
ROE_t				0.078	0.083	0.056	0.050	0.059	0.047	0.045	0.055	0.030	0.036	0.037
				(6.98)	(6.23)	(4.76)	(4.99)	(4.86)	(4.19)	(5.17)	(5.24)	(2.73)	(3.71)	(3.97)
E_t/P_t				-0.027	-0.028	-0.031	-0.012	-0.013	-0.014	-0.017	-0.006			
				(-3.25)	(-3.19)	(-3.22)	(-1.55)	(-1.76)	(-1.56)	(-1.77)	(-0.93)			
$ExpInv_t/P_t$							0.022	0.021	0.021	0.032	0.024		0.028	0.025
							(4.50)	(4.58)	(4.09)	(2.66)	(4.26)		(4.59)	(4.41)
$Beta_t$										0.008	-0.004			
										(0.75)	(-0.72)			
$Size_t$										-0.038	-0.002			
										(-1.14)	(-0.21)			
MOM_t										0.014	-0.008			
										(0.61)	(-1.79)			
Adjusted R^2	0.001	0.006	0.008	0.026	0.023	0.035	0.031	0.020	0.046	0.033	0.041	0.011	0.025	0.016

Panel C: Mean Annual Returns for Portfolios formed on CP_t/P_t, and OP_t/P_t

The High (Low) portfolio consists of the top (bottom) 1/3 of the ranked number each year.

	Negative CP	Positive CP	Negative OP	Positive OP
Low	0.082	0.081	0.102	0.094
Medium	0.047	0.125	0.068	0.118
High	0.040	0.181	0.046	0.163
High - Low	-0.042	0.100	-0.057	0.069
	(-1.48)	(3.64)	(-1.68)	(2.41)

Table 6: Mean Annual Returns for the SUMIS Factor and Component Portfolios

Panel A reports mean excess annual returns over years for portfolios formed by sorting firms each year on E_t/P_t and ROE_t. Panel B reports returns from sorting on E_t/P_t, ROE_t, and $ExpInv_t/P_t$ and for the SUMIS factor. High (Low) portfolios consist of the top (bottom) 1/3 of the ranked number each year. Portfolios are formed at March 31 each year, 1964-2018. Return differences between indicated portfolios are also reported, with t-statistics calculated as the mean return differences relative to their standard errors calculated from the time series of return differences. Variables are defined in the appendix. Firms with negative lagged book values are excluded.

Panel A: Mean Returns for Portfolios formed on E_t/P_t and ROE_t

		Positive Earnings			Diagonal Spread Return
	Negative Earnings	Low E/P	Medium E/P	High E/P	a – b
1: High ROE_t	0.138	0.080[b]	0.100	0.128	
2: Medium	0.131	0.087	0.101	0.139	
3: Low ROE_t	0.067	0.133	0.138	0.156[a]	
3 − 1	-0.071	0.053	0.038	0.028	0.076
	(-2.58)	(2.54)	(3.20)	(2.16)	(2.57)

Panel B: Mean Returns for Portfolios formed on E_t/P_t, ROE_t, and Expensed Investment ($ExpInv_t/P_t$)

		Positive Earnings			Diagonal Spread Return
	Negative Earnings	Low E/P	Medium E/P	High E/P	a – b (SUMIS)
1: High ROE_t and Low $ExpInv_t/P_t$	0.064	0.054[b]	0.064	0.098	
2: Medium	0.107	0.095	0.110	0.140	
3: Low ROE_t and High $ExpInv_t/P_t$	0.146	0.183	0.182	0.187[a]	
3 − 1	0.082	0.129	0.118	0.089	0.133
	(1.52)	(4.02)	(4.60)	(3.01)	(3.23)

Table 7: Mean Annual Returns for the SUMBS Factor and Component Portfolios

The table reports mean excess annual returns from sorting firms each year on the change in net operating assets relative to price ($\Delta NOA_t/P_{t-1}$) and financing leverage ($Debt_t/P_t$) and for the SUMBS factor. High (Low) portfolios consist of the top (bottom) 1/3 of the ranked number each year. Portfolios are formed at March 31 each year, 1964-2018. Return differences between indicated portfolios are also reported, with t-statistics calculated as the mean return differences relative to their standard errors calculated from the time series of return differences. Variables are defined in the appendix.

Mean Returns for Portfolios formed on $\Delta NOA_t/P_{t-1}$ and $Debt_t/P_t$

	$\Delta NOA_t/P_{t-1}$			Diagonal Spread Return a – b (SUMBS)
	Low	Medium	High	
1: Low $Debt_t/P_t$	0.141	0.103	0.080[b]	
2: Medium $Debt_t/P_t$	0.159	0.101	0.088	
3: High $Debt_t/P_t$	0.179[a]	0.129	0.098	
3 − 1	0.038	0.025	0.018	0.099
	(1.08)	(1.30)	(0.97)	(3.06)

Table 8: Mean Annual Returns for the SUMIS Factor and Component Portfolios Differentiated by Firm Size

Panel A reports median ln(Size) for the portfolios for forming SUMIS in Table 6. Panel B reports mean annual excess returns for the SUMIS portfolios when microcap stocks are excluded. Panel C also excludes microcaps but reports returns for the remaining (larger) firms that have the accounting characteristics that determine the SUMIS portfolios. Firms with negative lagged book values are excluded.

Panel A: Median ln(Market Capitalization) for Portfolios Underlying the SUMIS Factor

		Positive Earnings			Diagonal Spread
	Negative Earnings	Low E/P	Medium E/P	High E/P	a – b
1: High ROE_t and Low $ExpInv_t/P_t$	4.375	6.116[b]	5.961	5.238	
2: Medium	3.356	5.207	5.487	4.596	
3: Low ROE_t and High $ExpInv_t/P_t$	2.360	3.450	3.889	3.550[a]	
3 − 1	-2.015	-2.667	-2.071	-1.688	-2.566
	(-17.58)	(-32.05)	(-24.80)	(-16.17)	

Panel B: Mean Returns for Portfolios Underlying the SUMIS Factor, Excluding Microcaps

Microcap stocks are those below the 20th percentile of NYSE stocks by size.

		Positive Earnings			Diagonal Spread Return
	Negative Earnings	Low E/P	Medium E/P	High E/P	a – b
1: High ROE_t and Low $ExpInv_t/P_t$	0.064	0.056[b]	0.073	0.095	
2: Medium	0.055	0.083	0.097	0.116	
3: Low ROE_t and High $ExpInv_t/P_t$	0.157	0.149	0.127	0.143[a]	
3 − 1	0.093	0.093	0.054	0.048	0.086
	(1.73)	(3.17)	(3.03)	(1.71)	(2.14)

Panel C: Mean Returns for Portfolios Underlying the SUMIS Factor, Excluding Microcaps but with Returns in Rows 1 and 3 only for those Non-microcaps Satisfying the Cut-offs for Portfolios in Table 6

		Positive Earnings			Diagonal Spread Return a – b
	Negative Earnings	Low E/P	Medium E/P	High E/P	
1: High ROE_t and Low $ExpInv_t/P_t$	0.055	0.055[b]	0.073	0.094	
2: Medium	0.060	0.083	0.097	0.116	
3: Low ROE_t and High $ExpInv_t/P_t$	0.151	0.149	0.127	0.145[a]	
3 − 1	0.096	0.094	0.054	0.052	0.091
	(1.62)	(3.17)	(3.03)	(1.84)	(2.23)

Table 9: Tests of Accounting Factors Against Fama and French Factors

The table reports estimated intercepts, α, and slope coefficients, β, from time-series regressions of monthly excess returns for portfolios underlying SUMIS (in Panel B of Table 6) on Fama and French factor returns. In Panel A, the returns are for the high-risk, long portfolios for each E/P portfolio in the SUMIS factor (on row 3 in Panel B of Table 6) and in Panel B they are for the low-risk, short portfolios (on row 1). These are differentiated on ROE and expensed investment. Panel C is the return difference between the long and short portfolios (rows 3 – 1). The All portfolio in each panel is all positive E/P stocks pooled into one portfolio. The return in the last column in Panels A and B is that for the high positive E/P portfolio minus that low positive E/P portfolio in row 3 and row 1 respectively. In Panel C, it is that for SUMIS capturing the diagonal spread in Panel B of Table 6, a – b. Portfolios are reformed each March 31, 1964 to March 31, 2018, with the monthly returns running from April, 1964 to March, 2019. The Fama and French factor returns are from the Kenneth French web library at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Panel A: Long (High Risk) Portfolios (Low ROE_t, High $ExpInv_t/P_t$) for E/P Portfolios

	Negative Earnings	Positive Earnings				High E/P – Low E/P Portfolio
		Low E/P	Medium E/P	High E/P	All	
α	0.013	0.011	0.009	0.010	0.010	0.010
	(3.79)	(8.45)	(9.06)	(7.73)	(10.77)	(7.73)
β_{MKT}	0.987	0.959	0.967	1.005	0.976	1.005
	(11.56)	(29.32)	(38.96)	(32.88)	(42.77)	(32.88)
β_{HML}	0.114	0.308	0.360	0.605	0.419	0.605
	(0.67)	(4.76)	(7.33)	(10.00)	(9.27)	(10.00)
β_{SMB}	1.683	1.345	1.047	1.122	1.169	1.122
	(13.97)	(29.15)	(29.30)	(26.04)	(36.31)	(26.04)
β_{UMD}	-0.371	-0.235	-0.131	-0.319	-0.226	-0.319
	(-4.54)	(-7.50)	(-5.50)	(-10.91)	(-10.32)	(-10.91)
β_{RMW}	-1.197	0.076	0.166	0.296	0.177	0.296
	(-7.07)	(1.17)	(3.37)	(4.87)	(3.92)	(4.87)
β_{CMA}	-0.002	0.071	0.194	0.110	0.130	0.110
	(-0.01)	(0.76)	(2.71)	(1.25)	(1.98)	(1.25)
Adjusted R^2	0.55	0.81	0.85	0.81	0.88	0.81

Panel B: Short (Low Risk) Portfolios (High ROE_t, Low $ExpInv_t/P_t$) for E/P Portfolios

	Negative Earnings	Positive Earnings				
		Low E/P	Medium E/P	High E/P	All	High E/P – Low E/P Portfolio
α	0.005	0.002	0.002	0.003	0.002	0.002
	(2.80)	(2.30)	(1.99)	(2.50)	(3.12)	(2.30)
β_{MKT}	1.131	1.099	1.053	1.059	1.069	1.099
	(25.58)	(46.42)	(49.26)	(41.34)	(63.72)	(46.42)
β_{HML}	0.421	-0.339	0.111	0.444	0.077	-0.339
	(4.80)	(-7.23)	(2.62)	(8.75)	(2.30)	(-7.23)
β_{SMB}	1.115	0.603	0.682	0.836	0.705	0.603
	(17.87)	(18.06)	(22.63)	(23.14)	(29.80)	(18.06)
β_{UMD}	-0.113	0.004	-0.088	-0.240	-0.110	0.004
	(-2.67)	(0.16)	(-4.30)	(-9.77)	(-6.86)	(0.16)
β_{RMW}	-0.361	0.062	0.358	0.458	0.281	0.062
	(-4.11)	(1.32)	(8.44)	(9.01)	(8.43)	(1.32)
β_{CMA}	-0.208	-0.390	-0.118	-0.178	-0.234	-0.390
	(-1.64)	(-5.73)	(-1.93)	(-2.43)	(-4.86)	(-5.73)
Adjusted R^2	0.73	0.88	0.87	0.84	0.92	0.88

Panel C: Long minus Short Portfolios within E/P Portfolios

	Negative Earnings	Positive Earnings				SUMIS Portfolio:
		Low E/P	Medium E/P	High E/P	All	Diagonal Spread a-b
α	0.008	0.009	0.007	0.007	0.008	0.007
	(2.27)	(5.44)	(6.14)	(4.49)	(7.27)	(4.42)
β_{MKT}	-0.142	-0.139	-0.086	-0.054	-0.093	-0.094
	(-1.66)	(-3.42)	(-2.89)	(-1.41)	(-3.48)	(-2.28)
β_{HML}	-0.306	0.647	0.249	0.161	0.343	0.944
	(-1.79)	(8.01)	(4.24)	(2.12)	(6.50)	(11.58)
β_{SMB}	0.567	0.742	0.365	0.286	0.464	0.520
	(4.68)	(12.90)	(8.71)	(5.29)	(12.35)	(8.95)
β_{UMD}	-0.256	-0.239	-0.043	-0.080	-0.115	-0.323
	(-3.12)	(-6.11)	(-1.50)	(-2.16)	(-4.52)	(-8.19)
β_{RMW}	-0.835	0.014	-0.192	-0.163	-0.103	0.234
	(-4.90)	(0.17)	(-3.26)	(-2.13)	(-1.95)	(2.86)
β_{CMA}	0.207	0.461	0.312	0.288	0.364	0.500
	(0.84)	(3.93)	(3.66)	(2.62)	(4.76)	(4.23)
Adjusted R^2	0.12	0.41	0.25	0.11	0.39	0.49

Table 10: Accounting Factors and *q*-Investment Factors

The table reports estimated intercepts, α, and slope coefficients, β, from time-series regressions of monthly excess returns for portfolios underlying SUMIS (in Panel B of Table 6) on factor returns for *q*-investment models. As is Table 9, Panel A is for the high-risk, long portfolios for each E/P portfolio (on row 3 in Panel B of Table 6) and Panel B is for the low-risk, short portfolios (on row 1). These are differentiated on ROE and expensed investment. Panel C is the return difference between the long and short portfolios (rows 3 – 1). The All portfolio in each panel is all positive E/P stocks pooled into one portfolio. The return for the summary portfolio in Panels A and B is that for the high positive E/P portfolio minus that low positive E/P portfolio in row 3 and row 1 respectively. In Panel C, it is that for SUMIS capturing the diagonal spread in Panel B of Table 6, a – b. Portfolios are reformed each March 31, 1964 to March 31, 2018, with the monthly returns running from April, 1964 to March, 2019. The *q*-investment factor returns are at http://global-q.org/index.html.

Panel A: Long (High Risk) Portfolios (Low ROE_t, High $ExpInv_t/P_t$) for E/P Portfolios

	Negative Earnings	Positive Earnings				
		Low E/P	Medium E/P	High E/P	All	High E/P – Low E/P Portfolio
α	0.012	0.011	0.009	0.010	0.010	0.010
	(3.22)	(7.15)	(7.87)	(6.02)	(8.56)	(6.02)
β_{MKT}	1.085	0.970	0.945	1.002	0.971	1.002
	(13.41)	(27.93)	(35.27)	(27.84)	(36.98)	(27.84)
β_{IA}	-0.210	0.100	0.322	0.558	0.325	0.558
	(-1.14)	(1.26)	(5.26)	(6.80)	(5.42)	(6.80)
β_{ME}	0.015	0.012	0.010	0.010	0.010	0.010
	(13.66)	(25.00)	(26.31)	(19.70)	(28.96)	(19.70)
β_{ROE}	-1.514	-0.537	-0.285	-0.510	-0.440	-0.510
	(-10.29)	(-8.50)	(-5.85)	(-7.79)	(-9.22)	(-7.79)
β_{EG}	0.306	0.187	0.098	0.152	0.146	0.152
	(1.40)	(2.00)	(1.36)	(1.57)	(2.06)	(1.57)
Adjusted R^2	0.59	0.80	0.83	0.76	0.86	0.76

Panel B: Short (Low Risk) Portfolios (High ROE_t, Low $ExpInv_t/P_t$) for E/P Portfolios

	Negative Earnings	Positive Earnings				
		Low E/P	Medium E/P	High E/P	All	High E/P – Low E/P Portfolio
α	0.009	0.003	0.003	0.005	0.003	0.003
	(4.91)	(2.65)	(2.42)	(3.48)	(4.19)	(2.65)
β_{MKT}	1.100	1.106	1.024	1.010	1.045	1.106
	(26.56)	(44.10)	(44.54)	(32.65)	(57.10)	(44.10)
β_{IA}	0.064	-0.763	-0.126	0.101	-0.266	-0.763
	(0.68)	(-13.34)	(-2.40)	(1.43)	(-6.36)	(-13.34)
β_{ME}	0.010	0.006	0.006	0.007	0.006	0.006
	(16.83)	(16.18)	(19.16)	(16.11)	(24.38)	(16.18)
β_{ROE}	-0.542	0.269	0.142	-0.067	0.103	0.269
	(-7.20)	(5.89)	(3.40)	(-1.18)	(3.10)	(5.89)
β_{EG}	-0.252	-0.230	-0.152	-0.200	-0.198	-0.230
	(-2.26)	(-3.40)	(-2.45)	(-2.40)	(-4.01)	(-3.40)
Adjusted R^2	0.77	0.88	0.86	0.79	0.92	0.88

Panel C: Long minus Short Portfolios within E/P Portfolios

	Negative Earnings	Positive Earnings				SUMIS Portfolio: Diagonal Spread a-b
		Low E/P	Medium E/P	High E/P	All	
α	0.003	0.008	0.007	0.005	0.007	0.007
	(0.69)	(4.22)	(5.18)	(2.81)	(5.49)	(3.19)
β_{MKT}	-0.015	-0.136	-0.079	-0.008	-0.074	-0.105
	(-0.18)	(-3.15)	(-2.64)	(-0.21)	(-2.74)	(-2.23)
β_{IA}	-0.274	0.863	0.447	0.457	0.590	1.322
	(-1.45)	(8.78)	(6.54)	(5.18)	(9.59)	(12.33)
β_{ME}	0.006	0.006	0.004	0.003	0.004	0.004
	(4.91)	(10.73)	(8.82)	(5.45)	(11.65)	(6.45)
β_{ROE}	-0.971	-0.806	-0.427	-0.443	-0.543	-0.778
	(-6.42)	(-10.27)	(-7.84)	(-6.30)	(-11.07)	(-9.10)
β_{EG}	0.557	0.417	0.250	0.352	0.343	0.382
	(2.49)	(3.59)	(3.10)	(3.38)	(4.72)	(3.02)
Adjusted R^2	0.12	0.39	0.28	0.16	0.43	0.37

Table 11: Explaining Factor Returns with Accounting Factors

Panels A and B report estimated intercepts, α, and slope coefficients, β, from time-series regressions of monthly excess returns for selected zero-net-investment portfolios on the excess return on the market factor, MKT, and the summary accounting factor from the income statement and balance sheet, SUMIS and SUMBS. The market factor is the return on the CRSP value-weighted return index. The SUMIS factor captures the diagonal spread of returns in Panel B in Table 6, a – b, based on E/P, ROE, and expensed investment. The SUMBS factor is constructed from the diagonal spread of returns in Table 7 based on investment and financing leverage. Panel A and C portfolios involve the full range of observations, including loss firms. HML, SMB, and RMW in Panel B are Fama and French factors returns tabulated at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. *q*-ROE and *q*-EG are factors in *q*-investment models tabulated at http://global-q.org/index.html. Portfolios in Panels A and C, along with SUMIS and SUMBS, are reformed each March 31, 1964 to March 31, 2018, with the monthly returns running from April, 1964 to March, 2019. Panels C and D add the EG factor return.

Panel A: Dependent Variables are Returns for Portfolios in the Column Headings
High (low) portfolios consist of the top (bottom) 1/3 of stocks.

	High B/P– Low B/P	High TotB/P – Low TotB/P	High E/P – Low E/P	Small Size– Large Size	High ROE – Low ROE
α	0.001	0.002	-0.000	0.000	0.003
	(1.18)	(2.68)	(-0.30)	(0.08)	(1.78)
β_{MKT}	-0.104	-0.054	-0.101	0.063	-0.218
	(-6.98)	(-3.62)	(-7.16)	(1.78)	(-6.26)
β_{SUMIS}	0.276	0.296	0.261	0.219	-0.140
	(18.24)	(19.69)	(18.31)	(6.08)	(-4.00)
β_{SUMBS}	0.446	0.433	0.126	0.443	-0.198
	(18.86)	(18.37)	(5.62)	(7.87)	(-3.61)
Adjusted R^2	0.76	0.76	0.60	0.29	0.14

Panel B: Dependent Variables are Factor Returns from Extant Factor Models

	HML	SMB	RMW	*q*-ROE	*q*-EG
α	-0.001	0.000	0.003	0.008	0.009
	(-0.61)	(0.00)	(3.83)	(8.57)	(13.31)
β_{MKT}	-0.120	0.224	-0.121	-0.146	-0.193
	(-6.46)	(8.58)	(-6.36)	(-7.02)	(-12.47)
β_{SUMIS}	0.182	0.080	-0.034	-0.140	-0.003
	(9.70)	(3.05)	(-1.78)	(-6.42)	(-0.21)
β_{SUMBS}	0.265	0.016	0.037	-0.047	-0.010
	(9.03)	(0.39)	(1.25)	(-1.39)	(-0.39)
Adjusted R^2	0.46	0.11	0.05	0.17	0.20

Panel C: Panel A with the Expected Growth Factor, EG, Added

	High B/P– Low B/P	High TotB/P – Low TotB/P	High E/P – Low E/P	Small Size– Large Size	High ROE – Low ROE
α	0.000	0.001	-0.001	0.002	-0.004
	(0.24)	(1.73)	(-1.70)	(2.73)	(-2.08)
β_{MKT}	-0.088	-0.043	-0.075	-0.042	-0.089
	(-5.28)	(-2.55)	(-4.80)	(-1.05)	(-2.31)
β_{SUMIS}	0.279	0.306	0.268	0.235	-0.148
	(17.97)	(19.51)	(18.28)	(6.31)	(-4.11)
β_{SUMBS}	0.442	0.422	0.116	0.418	-0.176
	(18.34)	(17.33)	(5.09)	(7.25)	(-3.16)
β_{EG}	0.102	0.067	0.139	-0.536	0.695
	(2.64)	(1.73)	(3.83)	(-5.81)	(7.81)
Adjusted R^2	0.77	0.77	0.62	0.33	0.21

Panel D: Panel B with the Expected Growth Factor, EG, Added

	HML	SMB	RMW	*q*-ROE
α	-0.002	0.005	-0.001	0.002
	(-1.78)	(4.16)	(-0.97)	(1.93)
β_{MKT}	-0.099	0.108	-0.033	-0.016
	(-4.66)	(3.89)	(-1.62)	(-0.78)
β_{SUMIS}	0.188	0.093	-0.033	-0.137
	(9.50)	(3.59)	(-1.76)	(-7.29)
β_{SUMBS}	0.261	-0.005	0.047	-0.010
	(8.51)	(-0.11)	(1.60)	(-1.38)
β_{EG}	0.131	-0.605	0.466	0.675
	(2.68)	(-9.41)	(9.89)	(14.47)
Adjusted R^2	0.47	0.22	0.18	0.38